CHAPTER 1

Introduction to Biology

Definition of Biology:

Biology is a natural science concerned with the study of life and living organisms, including their structure, function, growth, origin, evolution, distribution, and taxonomy.

Biology is a vast subject.

Scope of Biology

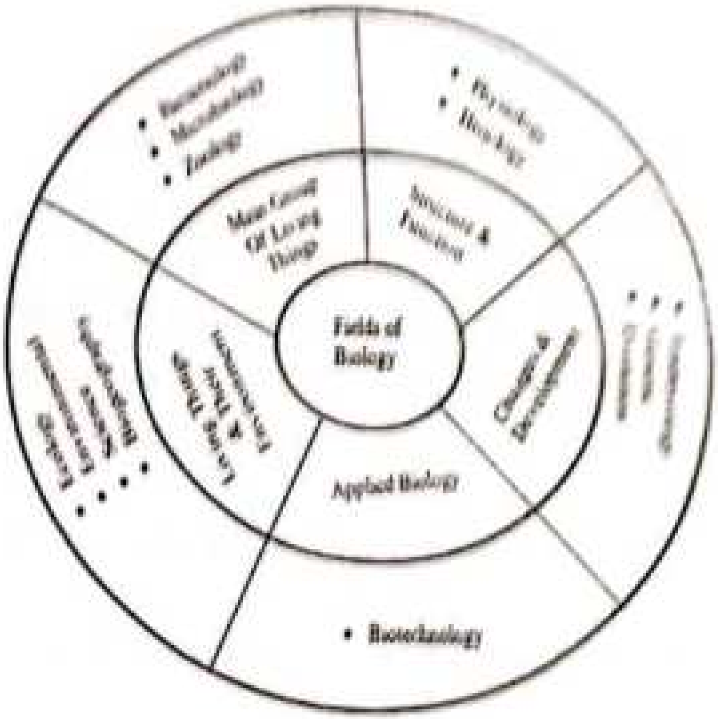


Figure 1: Scope of Biology

Biology:

The science which deals with the study of structure, organization, life processes, interactions, origin and evolution of living organisms is called **biology**.

Aristotle is regarded to as the **Father of Biology**

Biological science has been divided into 3 main branches:

- **Zoology:** It is the study of animals and the facts related to their life.
- **Botany:** It deals with the study of plants.
- **Microbiology:** It deals with the study of simpler, unicellular and microscopic organisms.

Biology reveals to us the secrets of life uncovered by biologists through centuries of researches.

It is of great importance to mankind in a practical sense and has various scopes, some of which are given below.

1. **Anthropology:**
   - The science of man and mankind including the study of the physical and mental constitution of man.
   - It also deals with the cultural development, social tradition as exhibited by them both in their past and present.

2. **Biomedical engineering:**
   - Branch of engineering dealing with the production of spare parts for man.
   - Biomedical engineers help in manufacturing of artificial limbs, heart, lungs etc. used by doctors to help impaired bodies function properly.

3. **Biotechnology:**
   - It deals with the use of living organisms or of substances obtained from them in industrial processes.

4. **Food Technology:**
   - The science of processing and preservation of healthy foods.
   - **Dairy technology:** The application of science for the manufacture of milk product.
   - **Apiculture:** The rearing of honey bees, bee keeping especially for commercial purposes.
   - **Fishery** or **Pisciculture:** The industry of rearing and catching fish or the products of the sea, lakes, rivers or ponds.
   - **Sericulture:** The breeding and treatment of silkworms for producing raw silk.

5. **Genetic Engineering:**
   - It involves genetic manipulations to produce an organism with a new combination of genes to improve the varieties.
   - Application of scientific knowledge to question civil and criminal laws is called "*forensic science*" which includes the study of finger prints, blood typing etc.

6. **Veterinary Medicine:**
   - It deals with the study of domesticated animals and their health care.
   - Science dealing with the rearing of domestic fowls such as chicken, ducks, turkeys etc. is called **poultry science**.

7. **Medicine:**
   - It is the science of treating diseases with drugs or curative substances.

- The science dealing with the study of nature of diseases, their causes, symptoms and effects is called **pathology**.
- The branch of medicine involving the physical operations to cure diseases or injuries to the body is called **surgery**.
- The science of knowledge of drugs and preparation of medicine or drugs is called **pharmacology**.
- Care of teeth, including cleaning and polishing, removal of spoiled teeth, filling and fitting of artificial teeth is called **dentistry**.

8. **Therapy:**
    - A method of treatment of convalescents and for physically handicapped people utilizing light work of diversion, physical exercise or vocational training is called **occupational therapy**.
    - The treatment of diseases, bodily weakness or defects by physical remedies such as massage and exercise is called **physiotherapy**.

Biology in Human Welfare

Health is defined as a state when an individual is free from illness and injury.

The Biology in Human Welfare comprises of the following three topics:

- *Human Health and Diseases*
- *Strategies for Enhancement of Food*
- *Microbes in Human Welfare*

Human Health and Diseases

Health is affected by different factors such as:

- Genetic disorders
- Infection caused by any bacteria, viruses, fungi etc.
- Life style of an individual such as food habits, sleeping habits, exercise etc.

There are various diseases that affect the Humans. The disease causing organism is known as **Pathogen**, Pathogen can be a *Bacterium, Virus, Fungi, and Nematode* etc. Some of the diseases with their causative agent and symptoms are given below-

Disease	Causative organism	Symptoms
Typhoid	*Salmonella typhi*	stomach pain, headache, loss of appetite, constipation.
Pneumonia	*Haemophilus influenza*	fever, coughs, chills etc.
Common cold	*Rhino* viruses	cough, headache, congestion, hoarseness, sore throat etc.

Ascariasis	*Ascaris lumbricoides*	internal bleeding, fever, muscle pain, intestinal blockage.
Filariasis	*Wuchereria bancrofti*	chronic inflammation in the lower limbs

What is Immunity?
Immunity is defined as the ability of a body to resist the infection.

What are different types of immunities?
There are two types of Immunities - **Innate Immunity** and **Adaptive Immunity.**
Innate Immunity: refers to the Non-Specific defence mechanism that works immediately when a person is encountered by an *Antigen*. For Example: Skin, Hydrochloric Acid in Stomach, White Blood Cells etc.

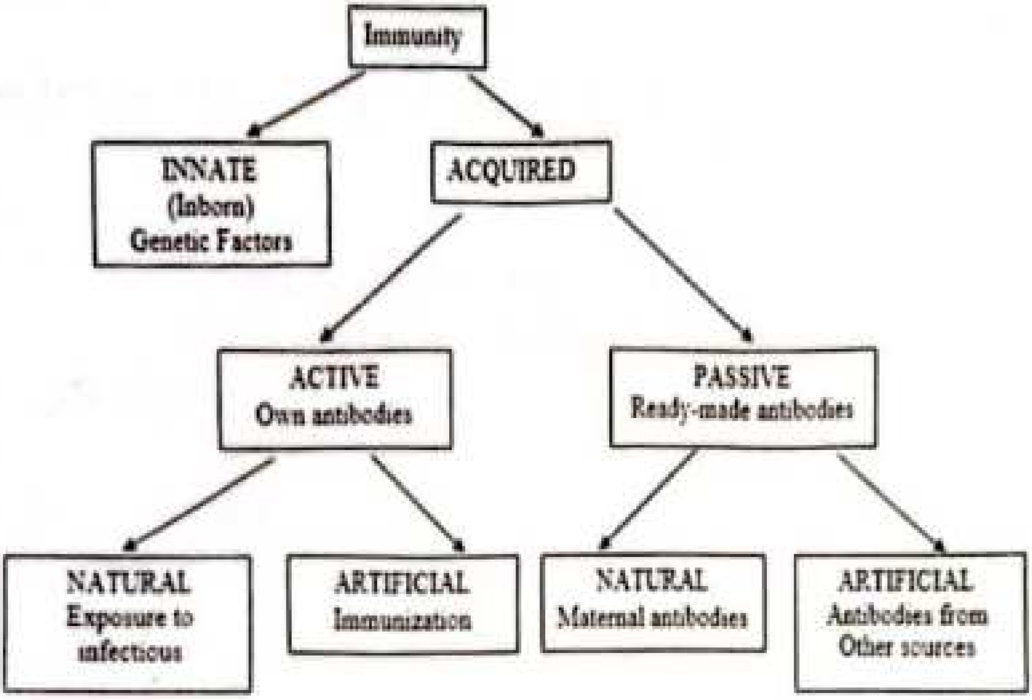


Figure 2: Different Types of Immunity.

Adaptive Immunity: refers to the specific defence mechanism. It involves two types of cells:
- **B Cells** which are formed in Bone Marrow.
- **T Cells** are formed in Bone Marrow but matures in Thymus.

Note: For detailed study, kindly refer to the topic **Human Health and Diseases.**

Strategies for Enhancement of Food
Biology also worked out in the enhancement of food production. Different mechanisms are employed to enhance the quality, quantity, texture, shelf life of different foods such as fruits, vegetables, milk and milk products etc.

Breeding is the most common methodology used for the enhancement of food production. Different methods of breeding are given below:

Cross Breeding | Outcrossing | Interspecific Hybridization | Inbreeding

Animal Husbandry is also another strategy that worked a lot for **Human Welfare**. It is a science of managing and caring of farm animals by human beings. It is a practice to raise livestock for promoting desirable traits in animals for **Human Welfare**. It includes the management of:

1. **Poultry Farm Management** (domesticated fowl for food such as meat and eggs)
2. **Dairy Farm Management** (management of the animals for milk, milk products)
3. **Bee Keeping** (maintenance of honey bees for the production of honey)
4. **Aquaculture/Pisciculture** (management of fishes)

Not only biology has helped in enhancement of animals but also plants welfare.
Plant breeding is the manipulation of plant species in order to obtain desirable plant traits such as more yield, pest resistance, insect resistance etc. This helps to produce disease resistant, insect/pest resistant varieties of different crops. This also enhances the yield of crops which is very much necessary for a country like India where there is a high demand of food.

Microbes in Human Welfare
Microbes are harmful as well as useful for human beings. Microbes brings lot of welfare for human beings:
Microbes in vaccination and antibiotics/microbes in industrial products

- Microbes in household products such as ***Lactobacillus*** in curd. This bacterium is essential for curd formation from milk. Cheese is also produced by fermented bacteria known as ***Propionibacterium Sharmanii.***
- Microbes also help in sewage treatment.
- Microbes are also used as bio control agents such as *Bacillus thuringiensis against insects/pests.*
- Some microbes are also used as bio-fertilizers. ***Rhizobium*** is a bacteria found in leguminous plants that helps in nitrogen fixation.
- There are certain microbes that colonize the human body such as *Streptococcus*, *Staphylococcus* etc. in stomach.

Strategies for Enhancement in Food Production
Characteristics of Life Concept in Biology
Every fall for the past dozen years, I have begun my biology class in much the same way---with a question. How do you know if something is alive? Now this may seem like a straightforward question, but based on student responses, it isn't nearly that easy to answer. Students often tell me that something is alive if it moves, or breathes, or thinks. Well, these responses are largely all wrong. However, despite the disappointment shown on aspiring

biology students' faces when they realize they can't yet distinguish between living and non-living, there is hope. Fortunately biologists have developed a list of eight **characteristics** shared by all living organisms. **Characteristics** are traits or qualities. Here is the list of characteristics shared by living things:

1. Cellular organization
2. Reproduction
3. Metabolism
4. Homeostasis
5. Heredity
6. Response to stimuli
7. Growth and development
8. Adaptation through evolution

let's take a moment and elaborate on each of these characteristics.

aracteristics of Life Defined

The first characteristic of life we listed was **cellular organization**. This simply means that living things are made of cells. Cells are the most basic unit of life. It doesn't matter if those cells are plants, animals, fungi, or bacteria. If something is going to be alive, it must be made of cells.

econd is **reproduction;** If something is alive it must be capable of reproducing. Multicellular life forms such as humans reproduce sexually, while unicellular life forms like bacteria reproduce asexually. The important thing to remember is that, in either case, living things reproduce.

Next, we come to our third characteristic, metabolism. This concept is a little difficult for some students to grasp. **Metabolism** is essentially a collection of chemical reactions occurring within the body (or cell). These reactions vary in form and function but promote processes such as protein synthesis, chemical digestion, cell division, or energy transformation. Because metabolism includes reactions that link to other characteristics, it is sometimes grouped with those other characteristics. However, for our purposes, we'll keep metabolism separate.

Our fourth characteristic is homeostasis. **Homeostasis** is the term used to describe maintaining a stable internal environment. In other words, think about how our bodies maintain a constant body temperature or how blood sugar levels are consistent. If homeostasis is disrupted, and we spike a fever, it's an indication that something is threatening life. The same holds true with blood sugar. When it gets too high or too low, homeostasis is disrupted, and unfortunately, this can be deadly. Therefore, maintaining homeostasis is a vital characteristic of life.

Next, we come to heredity. **Heredity** means that our genetic information can be passed from one generation to another. If either of your parents has dark eyes, and you also have dark eyes it's because of heredity.

Response to stimuli is the next characteristic in our countdown. This is simply a reaction to an internal or external force. This is something you've probably witnessed already. Think about a sunflower tilting towards the sun, a dog panting when hot, or trees losing their leaves in the fall when sunlight levels decrease. All living things respond to stimuli in some manner, to see it, all we have to do is look.

Topical Issues in Biology in biology: These are issues of current interest or relevance in biology. They are divided into three major groups as listed and discussed below:

1. General Biology
   1.1 Evolution and Origins of Life
   1.2 Biochemistry and Cell Biology
   1.3 Other

2. Non-human Biology
   2.1 Ecology, Evolution, and Palaeontology
   2.2 Ethology
   2.3 Non-Human Organs and Biomolecules

3. Neuroscience and Cognition
   3.1 Neurophysiology
   3.2 Cognition and Psychology

General Biology

Evolution and Origins of Life

- **Origin of life.** Exactly how and when did life on Earth originate? Which, if any, of the many hypotheses is correct?
  - **Origins of viruses.** Exactly how and when did different groups of viruses originate?
  - **Extra-terrestrial life.** Might life which does not originate from planet Earth also have developed on other planets? Might this life be intelligent?
  - What are the chemical **origins of life**? How did non-living chemical compounds generate self-replicating, complex life forms?
- **Evolution of sex.** What selective advantages drove the development of sexual reproduction, and how did it develop?
  - Homosexuality. What is the cause of homosexuality, especially in the human species?
  - **Development and evolution of brain.** How and why did the brain evolve? What are the molecular determinants of individual brain development?

Biochemistry and Cell Biology

- **What do all the unknown proteins do?** Almost two decades since the first eukaryotes were sequenced, the "biological role" of around 20% of proteins are still unknown. Many of these proteins are conserved across most eukaryotic species and some are conserved in bacteria, indicating a role fundamental for life.

- **Determinants of cell size:** How do cells determine what size to grow to before dividing?
- **Golgi apparatus:** In cell theory, what is the exact transport mechanism by which proteins travel through the Golgi apparatus?
- **Mechanism of action of drugs.** The mechanisms of action of many drugs including paracetamol, lithium, thalidomide and ketamine are not completely understood.
- **Protein folding:** What is the folding code? What is the folding mechanism? Can we predict the native structure of a protein from its amino acid sequence? Is it possible to predict the secondary, tertiary and quaternary structure of a polypeptide sequence based solely on the sequence and environmental information? Inverse protein-folding problem: Is it possible to design a polypeptide sequence which will adopt a given structure under certain environmental conditions? This has been achieved for several small globular proteins in recent years.
- **Enzyme kinetics:** Why do some enzymes exhibit faster-than-diffusion kinetics?[10]
- **RNA folding problem:** Is it possible to accurately predict the secondary, tertiary and quaternary structure of a poly-ribonucleic acid sequence based on its sequence and environment?
- **Protein design:** Is it possible to design highly active enzymes *de novo* for any desired reaction?
- **Biosynthesis:** Can desired molecules, natural products or otherwise, be produced in high yield through biosynthetic pathway manipulation?
- What is the *mechanism of allosteric transitions of proteins*? The concerted and sequential models have been hypothesised but neither has been verified.
- What are the **endogenous ligands of orphan receptors**?
- What substance is **endothelium-derived hyperpolarizing factor**?

Other

- **Why does biological aging occur?** There are a number of hypotheses why senescence occurs including those that it is programmed by gene expression changes and that it is the accumulative damage of biological processes.
- **Consistency of movement.** How can we move so controllably, even though the motor nerve impulses seem haphazard and unpredictable?
- **How do organs grow to the correct shape and size?** How are the final shape and size of organs so reliably formed? These processes are in part controlled by the Hippo signalling pathway

- **Can developing biological systems tell the time?** To an extent, this appears to be the case, as shown by the CLOCK gene.
- **Why are babies so rarely born with cancer?**

Non-Human Biology

Ecology, Evolution, and Palaeontology

Unsolved problems relating to the interactions between organisms and their distribution in the environment include:

- Paradox of the plankton. The high diversity of phytoplankton seems to violate the competitive exclusion principle.
- Cambrian explosion. What is the cause of the apparent rapid diversification of multicellular animal life around the beginning of the Cambrian, resulting in the emergence of almost all modern animal phyla?
- Latitudinal diversity gradient. Why does biodiversity increase when going from the poles towards the equator?
- Darwin's abominable mystery of botany/plants. What is the exact evolutionary history of flowers and what is the cause of the apparently sudden appearance of nearly modern flowers in the fossil record?
- Absence of Loricifera fossils. There are at least 100 species of this phylum of marine dwelling animals (many undescribed), but none of them is known to be present in the fossil record.
- Adult form of Facetotecta. The adult form of this animal has never been encountered in the water, and it remains a mystery to what it grows into.
- Origin of snakes. Did snakes evolve from burrowing lizards or aquatic lizards? There is evidence for both hypotheses.
- Origin of turtles. Did turtles evolve from anapsids or diapsids? There is evidence for both hypotheses.
- Ediacaran biota. How should Ediacaran biota be classified? Even what kingdom they belong to is unclear. Why were they so decisively displaced by Cambrian biota?

Ethology

Unsolved problems relating to the behaviour of animals include:

- Homing. A satisfactory explanation for the neurobiological mechanisms that allow homing in animals has yet to be found.
- Butterfly migration. How do the descendants of monarch butterfly all over Canada and the US eventually, after migrating for several generations, manage to return to a few relatively small overwintering spots?
- Blue whale. There is not much data on the sexuality of the blue whale.
- Gall wasp. It is largely unknown how gall wasps induce gall formation in plants; chemical, mechanical, and viral triggers have been discussed.

Non-Human Organs and Biomolecules

Unsolved problems relating to the structure and function of non-human organs, processes and biomolecules include:

- **Alkaloids**. The function of these substances in living organisms which produce them is not known
- Korarchaeota (archaea). The metabolic processes of this phylum of archaea are so far unclear.
- Rotifer. What is the function of the retrocerebral organ of rotifers (pseudocoelomate **animals**)?
- **Glycogen body**. The function of this structure in the spinal cord of birds is not known.
- **Arthropod head problem**. A long-standing zoological dispute concerning the segmental composition of the heads of the various arthropod groups, and how they are evolutionarily related to each other.
- Ovaries of **basking sharks**. Only the right ovary in female basking sharks appears to function. The reason is unknown.
- **Stegosaur** osteoderms/scutes. There is a long-standing debate over whether the primary function of the osteoderms/scutes of stegosaurs is protection from predators, sexual display, species recognition, thermoregulation, or other functions.

Neuroscience and Cognition

Neurophysiology

Sleep	What is the biological function of sleep? Why do we dream? What are the underlying brain mechanisms? What is its relation to anaesthesia?
Neuroplasticity	How plastic is the mature brain?
General anaesthetic	What is the mechanism by which it works?
Neuropsychiatric Diseases	What are the neural bases (causes) of mental diseases like psychotic disorders (e.g. mania, schizophrenia), Parkinson's disease, Alzheimer's disease, or addiction? Is it possible to recover loss of sensory or motor function?
Neural computation	What are all the different types of neuron and what do they do in the brain?

Cognition and Psychology

Cognition and Decisions	How and where does the brain evaluate reward value and effort (cost) to modulate behaviour? How does previous experience alter perception and behaviour? What are the genetic and environmental contributions to brain function?
Computational Neuroscience	How important is the precise timing of action potentials for information processing in the

	neocortex? Is there a canonical computation performed by cortical columns? How is information in the brain processed by the collective dynamics of large neuronal circuits? What level of simplification is suitable for a description of information processing in the brain? What is the neural code?
Computational Theory of Mind	What are the limits of understanding thinking as a form of computing?
Consciousness	What is the brain basis of subjective experience, cognition, wakefulness, alertness, arousal, and attention? Is there a "hard problem of consciousness"? If so, how is it solved? What, if any, is the function of consciousness?
Free will	Particularly the neuroscience of free will
Language	How is it implemented neutrally? What is the basis of semantic meaning?
Learning and memory	Where do our memories get stored and how are they retrieved again? How can learning be improved? What is the difference between explicit and implicit memories? What molecule is responsible for synaptic tagging?
Noogenesis - the emergence and evolution of intelligence	What are the laws and mechanisms - of new idea emergence (insight, creativity synthesis, intuition, decision-making, eureka); development (evolution) of an individual mind in the ontogenesis, etc.?
Perception	How does the brain transfer sensory information into coherent, private precepts? What are the rules by which perception is organized? What are the features/objects that constitute our perceptual experience of internal and external events? How are the senses integrated? What is the relationship between subjective experience and the physical world?

Career Opportunities in Biology:

What Can You Do With a Biology Degree?

Biological science is the study of life and is therefore one of the broadest subjects you can study. Biology encompasses everything from the molecular study of life processes right up to the study of animal and plant communities.

So, what can you do with a biology degree?

Biology degrees are extensive, so as you might expect, careers for biology graduates are equally as wide-ranging. Careers you could pursue with a biology degree include:

- Research Science
- Pharmacology
- Biology
- Ecology
- Nature Conservation
- Biotechnology
- Forensic Science
- Science Writing
- Teaching

Biology careers can lead you to study living organisms to help develop biological knowledge and understanding of living processes for a number of different purposes, including treatment of disease and sustaining the natural environment.

Many biology degree graduates choose to study at postgraduate level within a specialization or related field, in order to further their expertise and help career progression, although further study often isn't necessary for many.

Read on to find out more about the selection of typical – and less typical – biology careers available for both undergraduates and postgraduates.

Typical careers with a biology degree:

Biology Careers in Research

Scientific research is not only crucial within society but also a highly stimulating career for biology graduates. As a research biologist you'll aim to develop knowledge of the world around us by studying living organisms. Careers in research provide perhaps the broadest scope of all careers with a biology degree, as research can be conducted across all specializations.

Most common is research within the medical and life sciences, covering areas such as health and disease, neurology, genomics, microbiology and pharmacology. Researchers help to develop societal knowledge within many areas and, with the right additional qualifications, can be found within academia, research institutes, medical facilities and hospitals, and also within business and industry.

Biology Careers in Healthcare

Working in healthcare as a biologist will see you developing campaigns to help treat and cure illnesses such as AIDS, cancer, tuberculosis, heart disease, and many lesser-known illnesses and diseases. Although many roles are out of reach to students holding just an undergraduate degree (such as doctor and practitioner roles), the sector has a huge hiring capacity, and biologists are well sought-after in the medical world.

Healthcare biologists with the necessary qualifications and experience also work as veterinarians, doctors, nurses, dentists and other healthcare professionals. Biologists are recruited not only within hospitals and other medical facilities; they are also hired by organizations such as the Peace Corps in order to bring advanced healthcare to developing and war-torn regions.

Biology Careers in Environmental Conservation

As an environmental biologist you'll be interested in solving environmental problems and helping to protect natural resources and plant and animal wildlife to conserve and sustain them for future generations. Careers with a biology degree which fall under this remit include marine and/or aquatic biologist, zoo biologist, conservation biologist, ecologist and environmental manager. Biologists in these roles carry out recovery programs for endangered species and provide education for the general public. Hiring industries include charities and not-for-profit organizations, government and the public sector and ecological consultancies.

Biology Careers in Education

With a biology degree and a teaching qualification you'll be equipped to work within education. You'll enjoy working with young people and encouraging them to learn about the world, be that in a classroom, a lecture theatre, a laboratory or a museum. The higher up in the education world you go, the more qualifications you'll need; for instance, a university lecturer will often be required to have gained a master's degree or even a PhD, while a primary or secondary school teacher will usually only need an undergraduate degree and a teacher training qualification. If you do choose to undertake further study and go on to work within higher education, you may be able to produce your own research, have your work published and/or become a member of an advisory board within your field.

Less Typical Careers with a Biology Degree

A biology degree will equip you with many transferable skills that are sought-after in the workplace, whether that workplace is within a scientific industry or not.
Below is a selection of some of the less conventional careers you can pursue with a biology degree.

Careers in Biotechnology

Biotechnology is the use of scientific principles to develop and enhance technology within a number of sectors, including the consumer goods market, the technology market and business and industry. Focuses are often within agriculture, food science and medicine, where biotechnologists can be involved with genetic engineering, drug development and advancing medical technologies, such as nanotechnology.

Careers in Forensic Science

As a forensic scientist you'll be working within the legal sector, alongside police departments or law enforcement agencies, in order to test and process evidence gathered in criminal investigations. Many forensic scientists specialize in specific areas such as forensic odontology (dental evidence), forensic anthropology (the examination human of decomposition), crime scene examination and medical examiner roles (requiring further study).

Careers in Government and Policy

Biology careers in government will involve working closely with government officials and policy makers in order to advise on and create new legislation for growing topics such as biomedical research and environmental regulation. Your role will be to ensure that changes to the legal system are made based on solid science. You may work at regional or national level as a political advisor for scientific organizations and agencies or not-for-profit entities. You may also act as a representative for a political committee or group.

Careers in Business and Industry

The pharmaceutical sector is a multi-billion dollar industry and is in constant need of biologists to work in research and development and to test new products and prepare them for the marketplace. Other commercial industries where biologists may find roles include scientific services companies, marketing, sales and public relations.

Careers in Economics

If you have a strong numerical brain, you may want to go pursue a career in biological economics. This will require you to work within government or other organizations to examine the economic impact of biological problems on society, including such problems as extinction, deforestation and pollution. Related roles include socioeconomics (focused on humans), environmental economics (focused on preserving natural capital) and ecological economics (focused on the connection between natural ecosystems and human economies).

Careers in Science Publishing and Communication

Lastly, what can you do with a biology degree if you're also interested in the world of media? You might be surprised to discover that media and journalism careers with a biology degree are fairly wide-ranging as well. If you're interested in publishing or journalism you may want to use your biology degree to enter the industry as a science writer or working on a science publication such as a journal, magazine, website, TV program or film. Within these roles you'll be able to play a role in informing and educating the general public about biological issues that are becoming relevant in contemporary society.

CHAPTER 2

Diversity, Characteristics and Classification of Living Organisms

Characteristics of Living Organisms

There are the activities which make organisms different from non-living things:

- Movement
- Respiration
- Nutrition
- Sensitivity Irritability
- Growth
- Excretion
- Reproduction
- Death
- Adaptation
- Competition

Introduction to Diversity in Living Organisms

We all know that there are abundant of living organisms present on the earth. Many organisms are not identical to each other.

Biologist have identified and classified more than 1.7 million species of organisms on this earth. Most of these species are found in the tropical regions of the world.

Definition of Terms

- **Biodiversity** is a term used to describe the great variety of living organisms on Earth and their varied habitats.
- **Systematics** is the scientific study of biological diversity and its evolution.
- **Taxonomy** is a subdivision of systematic and it is defined as the science of biological classification. It deals with naming of organisms and their characterization. Taxonomy describes the relationship between items while classification simply groups the items. **Carolus Linnaeus** is known as the **Father of the Modern Taxonomy**.

- **Classification** is the orderly arrangement of organisms into groups based *on trait* similarity
- **What is evolution?** Over a course of time the living organisms accumulate changes. These changes could be in their body type or size or their features. These changes allow them to survive better with the change in environment. This is called "Evolution". The concept was introduced by Charles Darwin.
- **Primitive Organisms** are the ones that have an ancient or body design. Their bodies haven't undergone many changes with time. They are called **'Lower'** organisms as we
- **The Advanced Organisms** are those who have recently acquired body changes. They are also called as **'Higher'** organisms.

Classification of Living Things

Classification presented by Aristotle – He classified animals on the basis of their habitat land, water and air. But it can be easily observed that the animals that live at a particular hab say land are still so different from each other. Therefore, it was decided to classify the liv organisms on the basis of a hierarchy.

- Carolus Linnaeus (1707–1778) laid the foundations for modern biological nomencla now regulated by the Nomenclature Codes, in 1735.
- He distinguished two kingdoms of living things which are:
  - *Regnum Animale* (animal kingdom) and
  - *Regnum Vegetabile* (vegetable kingdom, for plants)
  - Linnaeus later included minerals in his classification system, placing them in a th kingdom called: *Regnum Lapideum*

This hierarchical classification was based on the similarities and dissimilarities in characteristics of the living organisms. Organisms having similar characteristics were plac in a similar category.

Why do we need to classify organisms?

1. If we classify organisms into several categories it will be easier for us to study them.
2. It will help us in understanding how these organisms evolve.
3. We can also understand how different organisms are related to each other.
4. We can learn why different organisms are found at distinct geographical conditions.
5. Classification systems provide relatively stable, unique, and unequivocal names organisms
6. It greatly improve our predictive powers
7. The major goal of the biological classification systems in regular use today is **to re evolutionary relationships**

The Order of Classification

- Kingdom
- Phylum / Division
- Class
- Order
- Family
- Genus
- Species

Species is called the **Basic Unit of Classification**. Species is a group of organism which can interbreed with each other. The picture below explains how humans are classified in a hierarchical order.

Hierarchy Classification - Formation of Kingdoms

Biologists categorized different organisms into several kingdoms

Linnaeus 1735 2 Kingdoms	Haeckel 1866[4] 3 Kingdoms	Chatton 1937[5] 2 empires	Copeland 1956[6] 4 kingdoms	Whittaker 1969[7] 5 kingdoms	Woese et al. 1977[8] 6 kingdoms	Woese et al. 1990[9] 3 domains
(not treated)	Protista	Prokaryota	Monera	Monera	Eubacteria	Bacteria
					Archaebacteria	Archaea
				Protista	Protista	
Vegetabilia	Plantae	Eukaryota	Protista	Fungi	Fungi	Eukarya
			Plantae	Plantae	Plantae	
Animalia	Animalia		Animalia	Animalia	Animalia	

Figure 1 -Formation of Kingdoms

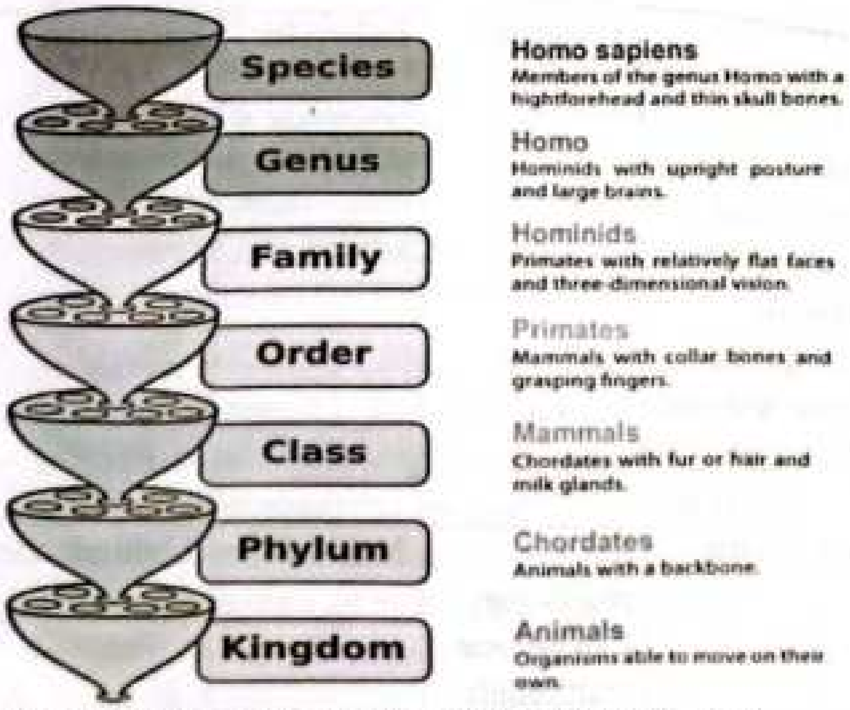


Figure 2 - Hierarchical Order of Classifying Humans

How scientists came up with the idea of kingdoms?

The scientists divided organisms into kingdoms on the basis of following criterion -

- **The organization inside the cells: Prokaryotic Cells** – Cells with no definite nucleus **Eukaryotic Cells** – Cells with a definite nucleus

  The organization of cells in the body: Unicellular – Single-celled organisms **Multicellular** – Multi-cell organisms.

  How organisms obtain their food: Autotrophs – Produce their food on their own **Heterotrophs** – Depend on other organisms for their food

Five Kingdom Classifications

The five kingdom system is the most common way of grouping organisms based on simple distinctive characteristics. Classification systems are always changing as new information is made available. A modern technology such as Genetics makes it possible to unravel evolutionary relationships to greater and greater detail. The five –kingdom system was developed by Robert H. Whittaker in 1969 and was built on the work of previous biologists such as Carolus Linnaeus.

Living Things can be classified into five major kingdoms:

- Kingdom Animalia
- Kingdom Plantae
- Kingdom Fungi
- Kingdom Protista
- Kingdom Monera (Bacteria)

Characteristics of the Organisms in Five-Kingdom Classification

	Monera	**Protista**	**Fungi**	**Plantae**	**Animalia**
Organization inside the cells	Consists of Prokaryotes.	Eukaryotes – some of them use appendages to move around such as flagella (whip-like structure) and Cilia (hair-like structure)	Eukaryotes	Eukaryotes	Eukaryotes
Organization of cells in the body	Unicellular	Unicellular	Initially unicellular. Can become multicellular	Multicellular	Multicellular

			in later stages of life		
Organisms obtain their food	Some of them are autotrophs like blue green algae while others are heterotrophs	Both autotrophs and heterotrophs	Heterotrophs. Most of them are decomposers or may be parasitic.	Autotrophs	Heterotrophs
Presence of cell wall	Some lack a cell wall while others have a cell wall	Only some have cell wall	Have cell walls. They are made up of complex sugar called chitin.	Have cell walls made of cellulose.	No cell walls
Example	Blue-green algae, Bacteria, Mycoplasma	Protozoan, Diatoms and Golden algae	Yeast and Mushroom (Agaricus), Rhizopus (Bread mould), Pencillium	Flowering plants, moss	Insects, reptiles

- Until the 1990's, five kingdoms were described: **Monera, Protista, Fungi, Plant, and Animal.**
- Then, in 2006, an entirely new classification was developed based upon a new discovery. **Dr.Carl Woese** and other scientists began to find evidence for a previously unknown group of prokaryotic organisms. These organisms lived in **extreme** environments - deep sea hydrothermal vents, "black smokers", hot springs, the Dead Sea, acid lakes, salt evaporation ponds - environments that scientists had never suspected would contain a profusion of life! Because they appeared prokaryotic, they were considered bacteria and named **"archaebacteria"** ('ancient' bacteria).
- However, it became obvious from biochemical characteristics and DNA and RNA sequence analyses that there were **numerous** differences between these archaebacteria and other bacteria. Before long, it was realized that these archaebacteria were **more closely related** to the eukaryotes (including ourselves!) than to bacteria. Today, these bacteria have been renamed **Archaea.** To accommodate the Archaea, systematists (specialists in taxonomy) devised an evolutionary model of classification with a level higher than a kingdom, called a domain.

- The three domains are **Archaea** (prokaryotes of extreme environments, like the archaebacteria), **Bacteria** (most of the known prokaryotes), and **Eukarya** (eukaryotes)

❖ The newest form of classification using the three domains includes many different kingdoms. The original **kingdom of Monera** was split between the **two domains Bacteria and Archaea**, while the domain **Eukarya includes** the previously described **kingdoms of Protista, Fungi, Plant, and Animal.**

Binomial System of Classification

- Binomial Nomenclature – is a formal system of naming species of living things by assigning two names to an organism i.e. Generic and specific names.
- Nomenclature is concerned with assignment of names in agreement with published rules
- The first word is the Generic name (genus) and it must start with capital letter
- The second word is the specific name (species) and it must starts with lower case
- The names must be italicized while printing (or underlined where italics is not possible)
- The scientific name can be abbreviated, where the genus is shortened to only its first letter followed by a period. E.g. *Homo sapiens* would become *H. sapiens*
- The genus name and species name should be underlined separately while writing
- Some examples of scientific name
- **Leo:** *Panthera leo*
- **Tiger:** *Panthera Tigris*
- **Human:** *Homo sapiens*
- **Mango:** *Mangifera indica*
- **Cowpea**: *Vigna unguiculata*

Why is nomenclature required?

It will help people identify an organism with a standard name anywhere in the world.

1. **The Kingdom Monera**
   - The Kingdom Monera includes organisms that are single-celled known as bacteria.
   - The microorganisms in Kingdom Monera are considered as the most ancient living forms on earth.
   - The kingdom is divided into two groups
     - *Archaebacteria* and
     - *Eubacteria*.
   - All the organisms of this kingdom are prokaryotes.

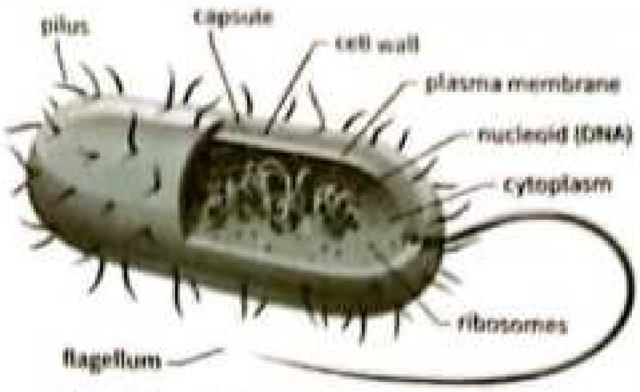


Figure 3- A Typical Prokaryote (Bacterium)

General characteristics of the kingdom Monera are as follows:

- They are **primitive** organisms
- All organisms of the kingdom are **prokaryotes**
- They are present in **both** living and non-living environment
- They can **survive** in harsh and extreme climatic conditions like in hot springs, acidic soils etc.
- They are **unicellular** organisms
- **Membrane bound** nucleus is **absent**
- **DNA** is in double stranded form, suspended in the cytoplasm of the organism, referred to as **nucleoid**
- A rigid **cell wall** is present
- Membrane bound cellular organelles like **mitochondria** are **absent**
- **Habitat** - they are found everywhere in hot springs, under ice, in deep ocean floor, in deserts and on or inside the body of plants and animals.
- **Nutrition** - occurs in various ways, some members of the groups could be:
  - ***autotrophs*** (ability to prepare their own food)
  - ***heterotrophs*** (depend on others for food)
  - ***saprophytes*** (feed on dead and decaying matter)
  - ***parasitic*** (live on other host cells for survival and cause)
- ***symbiotic*** - this is a mutual relationship with other organisms, which could be:
  - ***commensalism*** - it is where one organism is benefited and the other is not affected,
  - ***mutualism*** - where both the organisms are benefited.
- **Respiration** - respiration in these organisms vary, they may be
  - ***Obligate aerobes*** - the organisms must have oxygen for survival;
  - ***obligate anaerobes*** - the organisms cannot survive in the presence of oxygen;

- *facultative anaerobes* - these organisms can survive with or without oxygen

Circulation - is through diffusion

Movement - is with the help of flagella

Reproduction is mostly asexual, sexual reproduction is also seen.

- **Asexual reproduction** is by binary fission,
- **Sexual reproduction** is by conjugation, transformation and transduction.

assification of Kingdom Monera

e classification of kingdom Monera is based on the following features:

Classification based on habitat
Classification based on Shape
Classification Based on Mode of Nutrition
Classification based on Gram's staining

Classification Based on Habitat

- ***Archaebacteria:*** there are three major groups of bacteria based on their habitat:
  - Thermophiles
  - Halophiles
  - Methanogens

alophiles	Thermophiles	Methanogens
iese are salt loving cteria. They live in tremely salty water.	They live in boiling water such as hot springs and volcanoes.	They are found in the guts animals like cow and sheep. T produce methane gas from t dung.

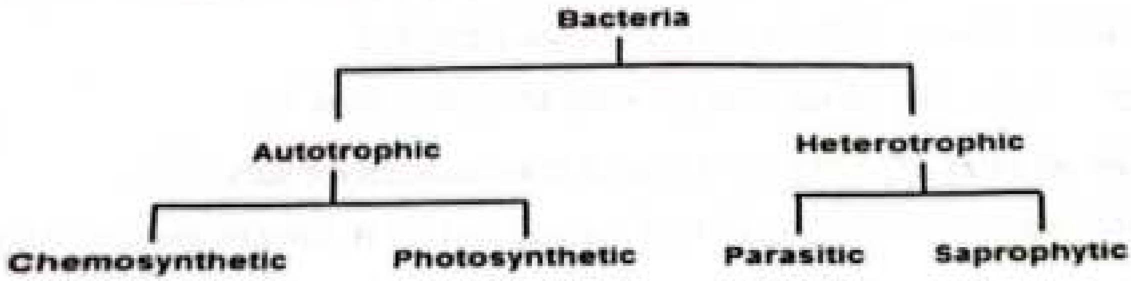
- ***Eubacteria*** *: these* are true bacteria

Classification Based on Shape

Bacteria can be classified into four groups based on shape:

- Spherical or round shaped bacteria (cocci)
- Rod-shaped (bacilli)
- Comma-shaped bacteria (vibrio)
- spiral shaped bacteria (spirilla)

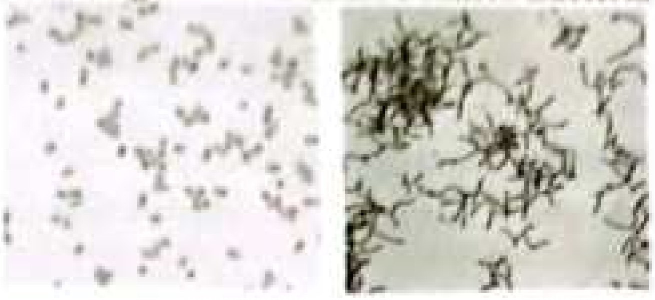
ssification Based on Mode of Nutrition



4. **Classification Based on Gram's Staining**

- **Gram's staining** is a test on cell walls developed by **Hans Christian Gram**
- This method helps to classify bacteria into:
  - ***Gram Positive Bacteria*** - The bacteria's cell wall is made up of protein-sugar complex that takes on purple colour during gram staining
  - ***Gram Negative Bacteria*** - The gram negative bacteria has an extra layer of lipid on the outside of the cell wall and appear pink during the Gram staining procedure

Gram Negative and Gram Positive Bacteria



Differences between Prokaryotes and Eukaryotes

Characteristic	Prokaryotes	Eukaryotes
Size of Cell	Typically 0.2-2.0 mm in diameter	Typically 10-100 mm in diameter
Nucleus	No nuclear membrane or nucleoli (nucleoid)	True nucleus, consisting of nuclear membrane & nucleoli
Membrane-Enclosed Organelles	Absent	Present; e.g. lysosomes, Golgi complex, endoplasmic reticulum, mitochondria & chloroplasts
Flagella	Consist of two protein building blocks	Complex; consist of multiple microtubules
Glycocalyx	Present as a capsule or slime layer	Present in some cells that lack a cell wall
Cell Wall	Usually present; chemically complex (typical bacterial cell wall includes peptidoglycan)	When present, chemically simple
Plasma Membrane	No carbohydrates and generally lacks sterols	Sterols and carbohydrates that serve as receptors present
Cytoplasm	No cytoskeleton or cytoplasmic streaming	Cytoskeleton; cytoplasmic streaming
Ribosomes	Smaller size (70S)	Larger size (80S); smaller size (70S) in organelles

Chromosome (DNA) arrangement	Single circular chromosome; lacks histones	Multiple linear chromosomes with histones
Cell Division	Binary fission	Mitosis
Sexual Reproduction	No meiosis; transfer of DNA fragments only (conjugation	Involves meiosis

2. The Kingdom Protista 1866

- The term Protista was first used by Ernst Haeckel in the year 1886.
- Kingdom Protista is a diverse group of eukaryotic organisms.
- Protists are unicellular, some are colonial or multicellular, they do not have specialize tissue organization.
- The simple cellular organization distinguishes the protists from other eukaryotes.
- The cell body of the protists contain a nucleus which is well defined and membran bound organelles
- Some have flagella or cilia for locomotion
- Reproduction in protists is both asexual and sexual.
- They live in any environment that contains water

All single celled organisms are placed under the Kingdom Protista

Characteristics of Kingdom Protista

- They are simple eukaryotic organisms.
- Most of the organisms are unicellular, some are colonial and some are multicellular li algae.
- Most of the protists live in water, some in moist soil or even the body of human and plant
- These organisms are eukaryotic, since they have a membrane bound nucleus and end membrane systems.
- They have mitochondria for cellular respiration and some have chloroplasts fo photosynthesis.
- Nuclei of protists contain multiple DNA strands; the number of nucleotides is significantl less than complex eukaryotes.
- **Movement** is often by flagella or cilia.
- **Respiration** - cellular respiration is primarily **aerobic** process, but some living in mu below ponds or in digestive tracts of animals are strictly **facultative anaerobes**.

Nutrition - they can be both **heterotrophic** and **autotrophic**.

Flagellates are filter feeding, some protists feed by the process of endocytosis (formatio of food vacuole by engulfing a bacteria and extending their cell membrane).

- **Reproduction** - some species have complex life cycle involving multiple organisms. Example: Plasmodium. Some reproduce sexually and others asexually.
- They can reproduce by **mitosis** and some are capable of **meiosis** for sexual reproduction.
- They form cysts in adverse conditions.
- Some protists are **pathogens** of both animals and plants. Example: ***Plasmodium falciparum*** causes malaria in humans.
- Protists are major component of plankton.

Classification of Kingdom Protista

Kingdom Protista are categorized into two taxon:

- *Protozoans - animal-like single-celled organisms:* They differ from animals; being unicellular while animals are multicellular.
- *Algae - plant-like single or multi-celled organisms.*

PROTOZOANS: Animal-like Protists

Protozoans are classified on the way they move into four categories:

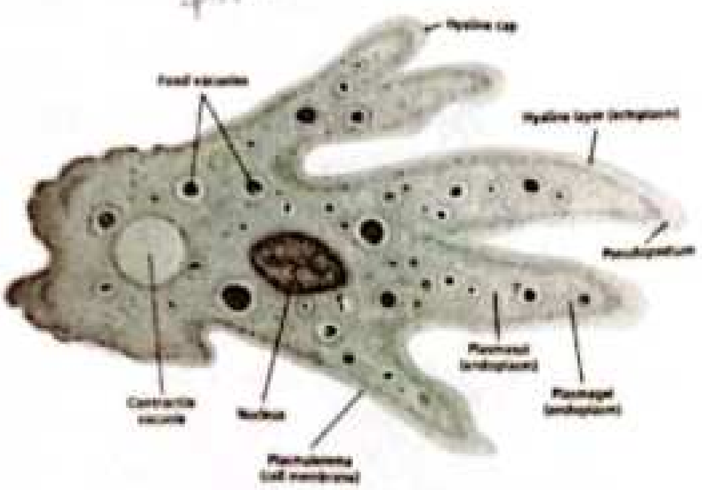
1. Phylum *Sarcodina*: move using pseudopod. Examples are *Amoeba, Foraminifera.*
2. Phylum *Mastigophora* (Zooflagellata): move using flagella. Examples are: *Trypanosoma gambiense*, causes sleeping sickness in cattle and human and *Trichonympha sp.*
3. Phylum *Ciliophora* (Ciliates): move using cilia. Examples are: *Paramecium and Vorticella*
4. Phylum *Sporozoa*: They forms spores and hence the name sporozoa. All members of this phylum are non-motile and parasitic. Example is *Plasmodium* - this parasite causes malaria in humans.

Some Common Examples of Protozoan

mitosis -> two daughter cells having same number of chromosomes as the parent cells

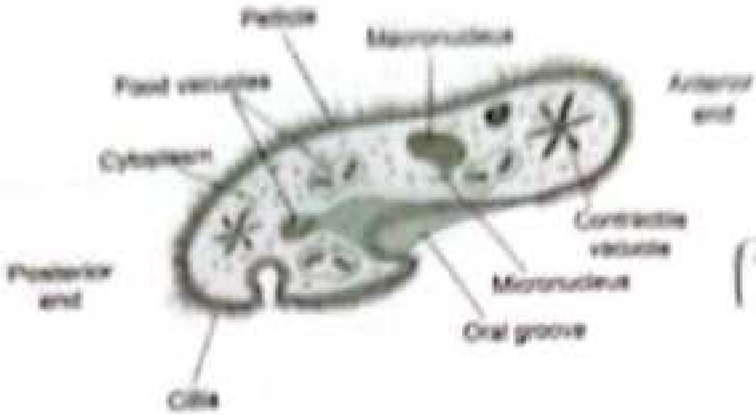
i.

Amoeba



meiosis -> reduce the number of Chromosomes in the parent cell by half and produce four gamete cells.

ii.



Paramecium

Figure 5: (i) *Amoeba*, (ii) *Paramecium*

ALGAE: Plant-like Protists

Plant-like protists have chlorophyll like that in plants. The green substances in their cells enable them to make food through photosynthesis.

They are classified as follows:

1. Phylum *Chlorophyta* (Green Algae). Example: *Spirogyra, Ulva, Chlamydomonas, Volvox*
2. Phylum *Rhodophyta* (Red Algae) Examples are: 'Nori' and Gelidium are used as food in parts of Asia. Carragean and agar are glue-like substances in red-algae. More examples are: *Porphyra, Rotalgen.*
3. Phylum *Phaeophyta* (Brown Algae). Example of rockweed is *Sargassum*. Algin is a substance derived from some algae which is used in making ice cream, lotion and plastics. More examples are: *Laminaria, Nereocystis*
4. Phylum *Chrysophyta* (Golden algae). There are three types of golden-algae:
   i. yellow-green algae,
   ii. golden brown algae,
   iii. diatoms.
5. Phylum *Pyrrophyta* (Fire Algae). Also called Phylum Dinoflagellata e.g. *Ceratium, Gonyaulax.*

Fungus-like Protists- Slime moulds, Water moulds e.g. *Saprolegnia*

Some common examples of Algae; green algae, red algae, brown algae, golden algae, fire algae and slime molds

3. The Kingdom Fungi

- Fungi are heterotrophic organisms; they obtain their nutrients by absorption
- The 'fruit' body of fungus is only seen, while the Vegetative body of the fungus is a mycelium.
- Mycelium is made up of tiny filaments called hyphae
- The organisms in kingdom fungi include: mushrooms, yeasts, molds, rusts, smuts, puffballs, truffles, morels, and molds
- More than 70,000 species of fungi have been identified
- Mycology is a discipline of biology which deals with the study of fungi.

Fungi are not capable of producing their own food, so they get their nourishment from other sources

Characteristics of Kingdom Fungi

- Fungi are eukaryotic organisms.
- They are non-vascular organisms.
- They reproduce by means of spores.
- Depending on the species and conditions both sexual and asexual spores may be produced.
- They are typically non-motile.
- Fungi exhibit the phenomenon of alteration of generation.
- The vegetative body of the fungi may be unicellular or composed of microscopic threads called hyphae.
- The structure of cell wall is similar to plants but chemically the fungi cell wall are composed of chitin.
- Fungi are heterotrophic organisms.
- They fungi digest the food first and then ingest the food, to accomplish this, fungi produce exo-enzymes (Extracellular digestion).
- Fungi store their food as **starch**.
- Biosynthesis of chitin occurs in fungi.
- The **nuclei** of the fungi are very small.
- During mitosis the **nuclear envelope** is not dissolved.
- **Nutrition** in fungi are either saprophytes, parasites or symbionts.
- **Reproduction** in fungi is both by sexual and asexual means.
  - Sexual state is referred to as **teleomorph**,
  - Asexual state is referred to as **anamorph**.

Some varieties of fungi; yeast (ascomycetes) smut in maize, mushroom

Figure 6: Mushroom

4. **Kingdom Plantae**

The criteria of classification in Plantae:

- Components of Plants – whether they are distinct or not
- Presence of Special Tissues in plants for the transportation of food and water
- Presence of Seeds – whether the seeds are present inside the fruits or not

Classification of plants on the ability to produce seeds:-

- **Cryptogams** – These plants do not have well developed reproductive organs. The org cannot be seen clearly as well and appear as if they are hidden. **Examples** are Thallophy Bryophyta and Pteridophyta.
- **Phanerogams** – These plants have well developed reproductive organs hence they produce seeds. They are further classified as the ones which have seeds hidden inside fr or not - **Gymnosperms** and **Angiosperms.**

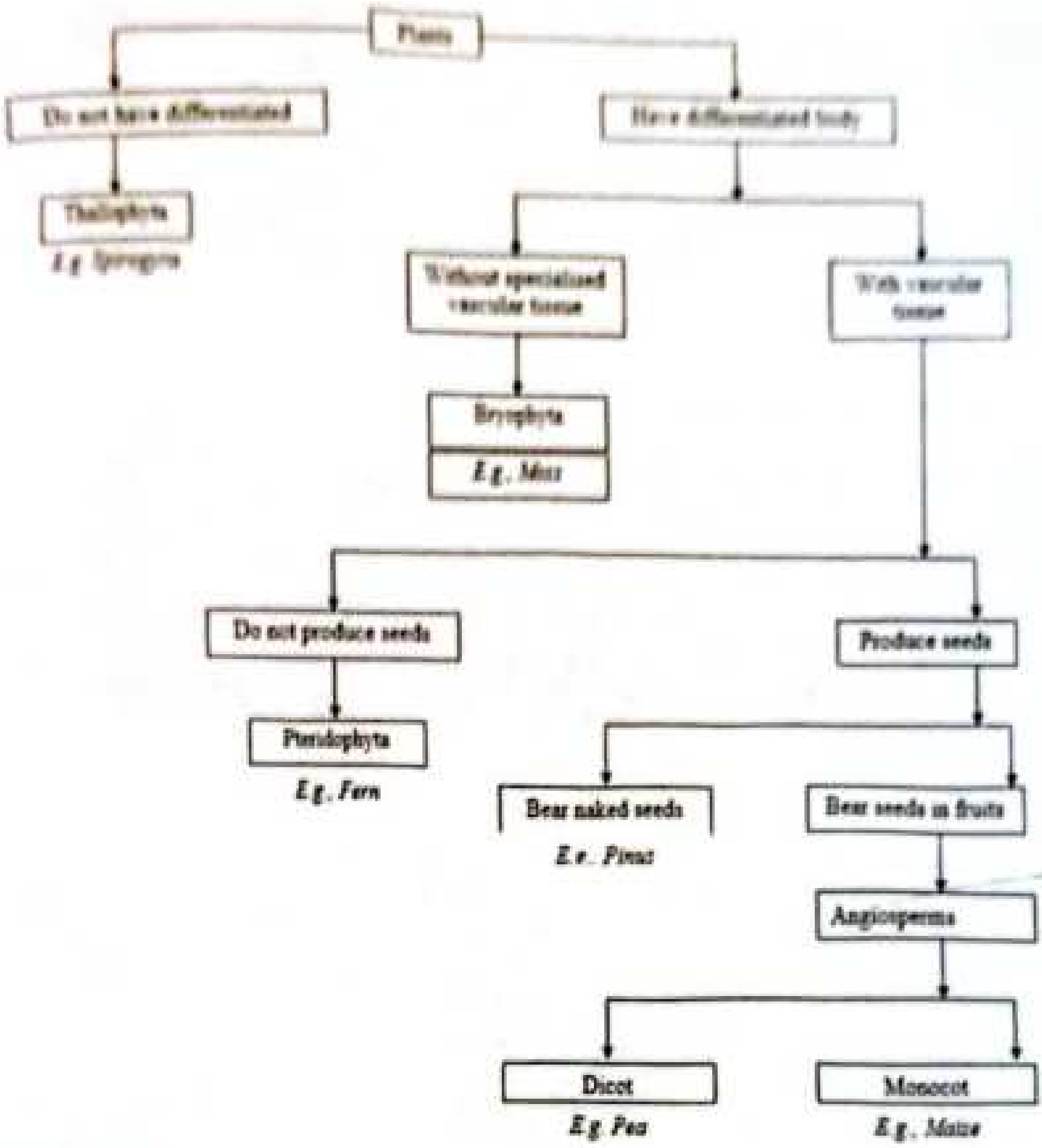
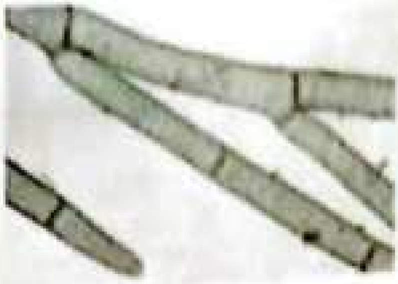


Figure 3: Plant Kingdom

Criteria	Thallophyta	Bryophyta	Pteridophyta
Components of plants	No distinct components. Undifferentiated Body	Little differentiated body. Distinct components are present as leaves and stem	Distinct components are present as roots, leaves and stem
Presence of special tissues- Vascular tissue	No	No	Yes
Presence of seeds	No	No	No
Found in	Aquatic environment, snow	First terrestrial plants but need water for sexual reproduction. So called as Amphibian of plant kingdom.	Terrestrial or dry areas
Example	Spirogyra, Ulothrix, Volvox	Moss and liverworts	Ferns


Ulothrix


Cladophora


Chara

Figure 8: Examples of Thallophyta


Liverwort


Hornwort


Moss

Figure 9 - Examples of Bryophyta


Maesilea


Fern

Figure 10 - Examples of Pteridophyta

Table 5: Difference between Gymnosperms and Angiosperms

	Gymnosperms	Angiosperms
The ability to produce seeds	Naked seeds	Seeds develop in an organ which th turns into the fruit
Existence	Exist for long time periods, Evergreen	Grow for varied time periods
Type	Woody, No flowers	Flowering plants
Meaning	Gymno – naked Sperm – seeds	Angio – Covered Sperma – seeds
Example	Pines, Deodar	Mustard, Maize, cassava, cowpea

What are Cotyledons?

The seed leaves in Angiosperms are called **Cotyledons**. They can grow on the germination of the seeds. Angiosperms can be divided into two types on the basis of the presence of cotyledons in them: Monocotyledons or Monocots, Dicotyledons or Dicots.

Criteria	Monocotyledons or Monocots	Dicotyledons or Dicots
Cotyledons (Seed Leaves)	Single Cotyledon	Two Cotyledons
Leaves	Long leaves, with parallel veins	Broad leaves with network of veins
Roots	Fibrous	Long taproot
Floral Parts	Multiples of three	Multiples of four or five
Example	Corn, Wheat, Grass	Rose, Sunflower, Lily

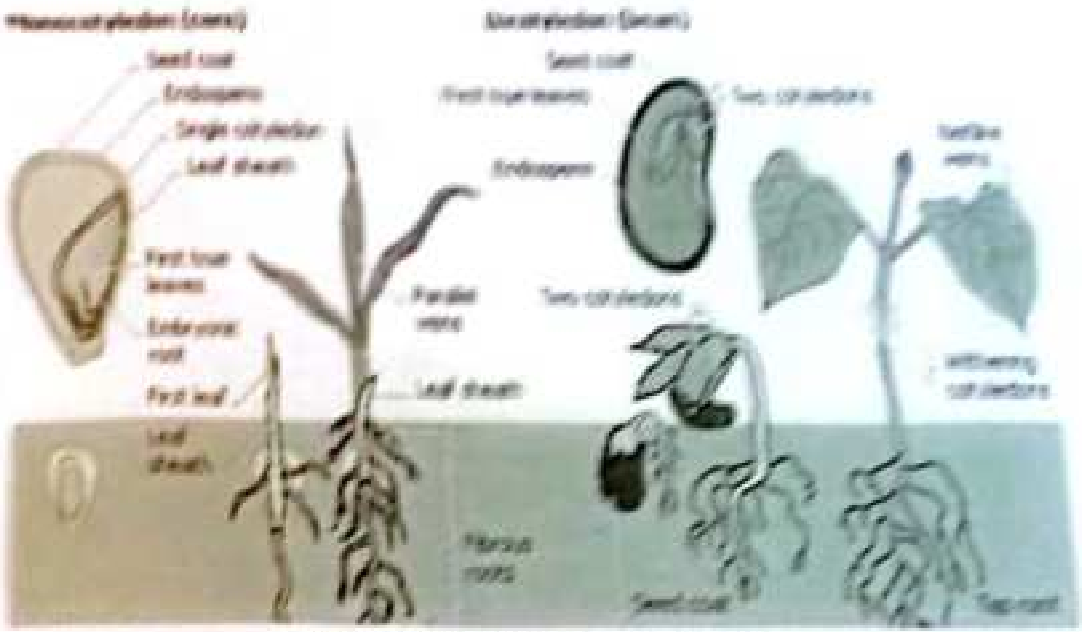


Figure 11: Monocots vs Dicots

5. Kingdom Animalia

Basic Characteristics of the Animalia Kingdom

i. Animals are eukaryotic, multicellular organisms that lack a cell wall.

ii. They are heterotrophs therefore they rely on others for food.

iii. They have a growth pattern. The adult animals have a specific shape and size.

iv. Most of the organisms have well-defined organ systems such as **Respiratory Sy** **Digestion System** and so on.

v. Most of the animals can move. They aren't stationary as **Plants**.

vi. Animals have a nervous system which is why they are able to respond to an e stimulus.

Animals are classified on the basis of differences in their body type and design. The body cavity or coelom in animals contains the organs. Based on the presence of body cavity animals can be categorized as:

i. **Coelomate** – They have true body cavity called **Coelom**
ii. **Pseudocoelomate** – It means false cavity. They have a body cavity which is filled with fluid
iii. **Acoelomate:** They have no body cavity at all.

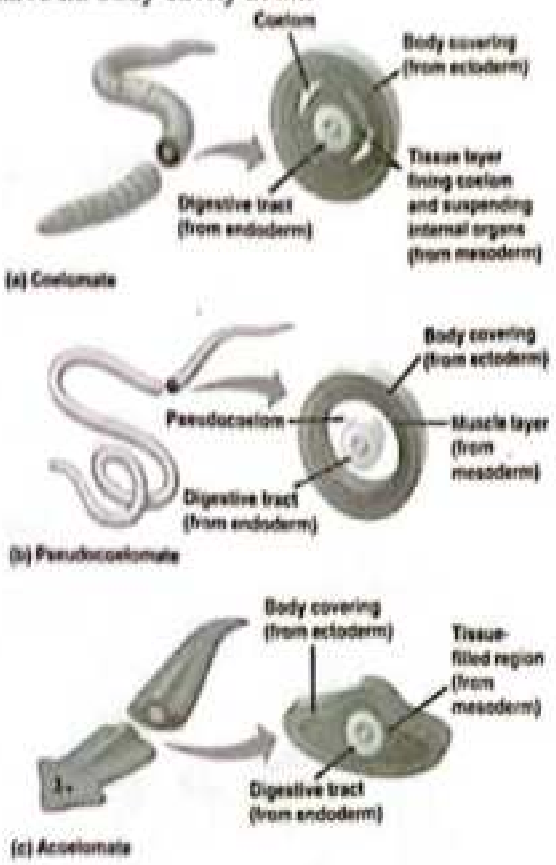
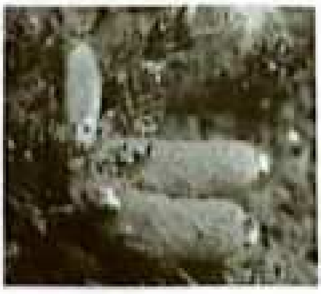


Figure 12

1. Phylum- Porifera

Spongilla



Sycon

Figure 13 - Phylum- Porifera

the wory body of a Porifera is called Ostia or Osculum

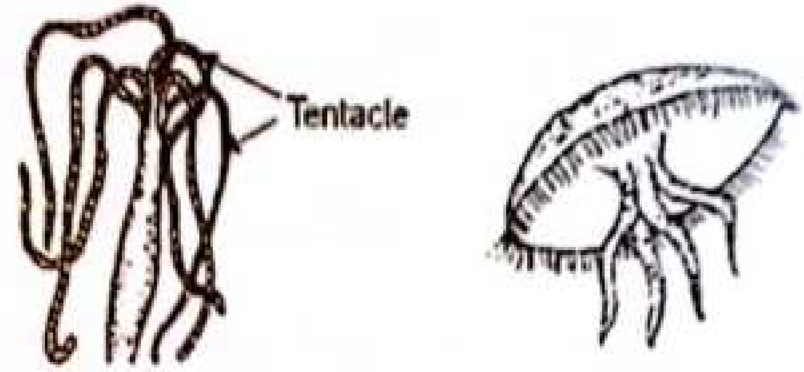
Characteristics of Phylum Porifera

- Level of Organization – Cells are present

what

- Symmetry – Asymmetrical
- Segmentation – No segments
- Body Cavity/ Coelom – No
- Presence of Organs – No
- Examples – Sycon, Spongilla, Euspongia
- They cannot move and are attached to a support.
- They have pores in their body
- These pores form a Canal system through which water and food circulate in the body and waste is removed.
- They have a skeleton made of sponging protein and calcium carbonate – hard covering on them

2. **Phylum- Coelenterata**



Hydra Jellyfish

Coelenterata

Figure 14 - Phylum- Coelenterata

Characteristics of Phylum Coelenterata Or Cya Cnid~~aria~~erian

- Level of Organization – Tissues, Cells have two layers – so called as **Dipinblastic Organism**
- Symmetry – Radial
- Segmentation – No segments
- Body Cavity/ Coelom – No
- Presence of Organs – No
- Examples – Aurelia (Jelly fish) and Adamsia (Sea Anemone)
- Other Characteristics –
- Some of them live in colonies - They are physically attached to each other such as **Cora**
- Some of them live solitary such as **Hydra**

* They are the sympist form of invertebrate
* The body of a cniderian is divided into two [illegible]

3. Phylum Platyhelminthes

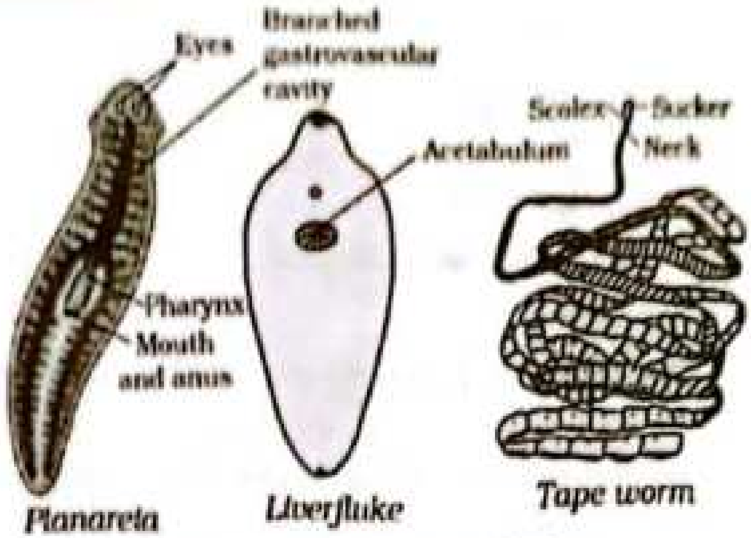


Figure 15: Phylum Platyhelminthes

Characteristics of Phylum Platyhelminthes

- Level of Organization – Organs. The cells have three layers – so are called **Triploblastic**
- Symmetry – Bilaterally Symmetrical - Left half of the body is identical to the right half
- Segmentation – No segments
- Body Cavity/ Coelom – No so called as **Acoelomates**
- Presence of Organs – Yes
- Examples – Taenia solium (Tapeworm), Fasciola hepatica (Liver Fluke)
- They have a flat body and thus are called **Flatworms**
- They can be Free-living like Planaria or parasitic. [Parasitic effect of is the presence of hook & sucker]

4. Phylum Nematoda

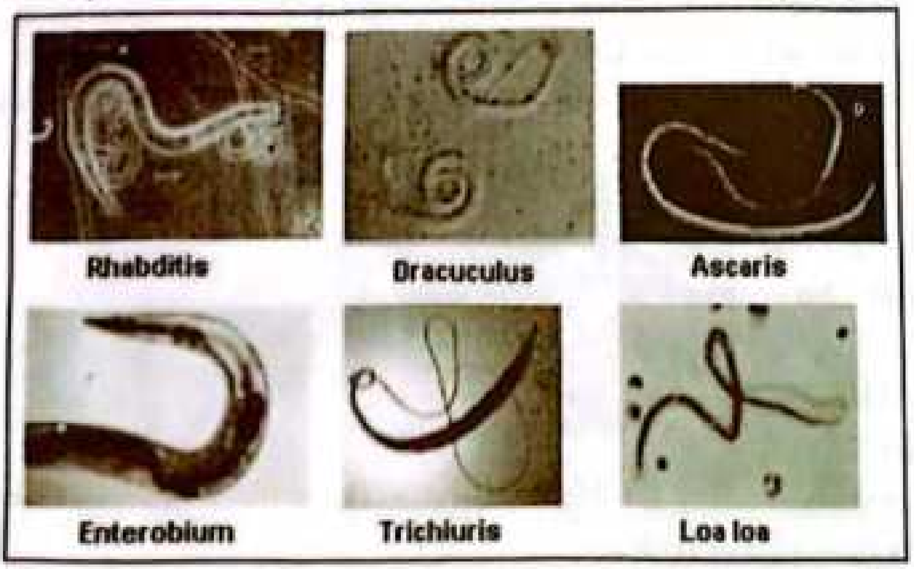


Figure 16: Phylum Nematoda

Characteristics of Phylum Nematoda

- Level of Organization – Tissues so are called **Triploblastic**
- Symmetry – Bilaterally Symmetrical - Left half of the body is identical to the right half
- Segmentation – No segments
- Body Cavity/ Coelom - False body cavity so called as **Pseudocoelomates**
- Presence of Organs – Organ System Level Organisation
- Examples – Parasitic worms and worms in the intestine
- They are called as **Round Worms**.
- Sexual dimorphism visible - Female and male worms are distinct.

5. Phylum Annelida

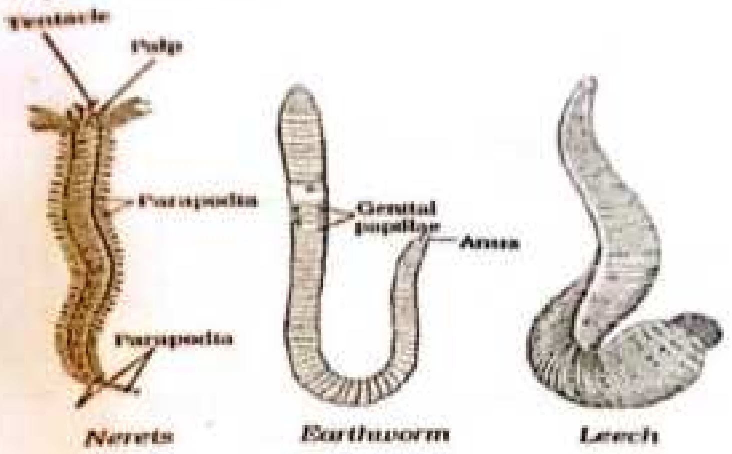


Figure 17: Phylum Annelida

Characteristics of Phylum Annelida

- Level of Organization – Organ system level, the cells have three layers so called **Triploblastic**
- Symmetry – Bilaterally Symmetrical
- True Segmentation – Present (organs can be identified separately)
- Body Cavity/ Coelom – True body cavity so called as **Coelomates**
- Presence of Organs – Definite organs
- Examples – Leech, Earthworms
- They are found in freshwater and marine water.
- They have closed Circulatory system.
- Nephridia is the excretory organ of Phylum Annelis
- they move with setae
- they are anaphrodite
- Annelida are metramerically symmetrical

6. Phylum Arthropoda

Figure 18: Phylum Arthropoda

Characteristics of Phylum Arthropoda

- Level of Organization – Organ systems
- Symmetry – Bilaterally symmetrical
- Segmentation – Present (organs can be identified separately)
- Body Cavity/ Coelom – True body cavity
- Presence of Organs – Definite organs
- Examples – Prawns and butterflies
- They have jointed legs
- They have an open circulatory system – There are no well-defined blood vessels
- They have chitinous exoskeleton
- Chitin is made up of calcium carbonate
- they respire through their face or trachea

7. Phylum Mollusca

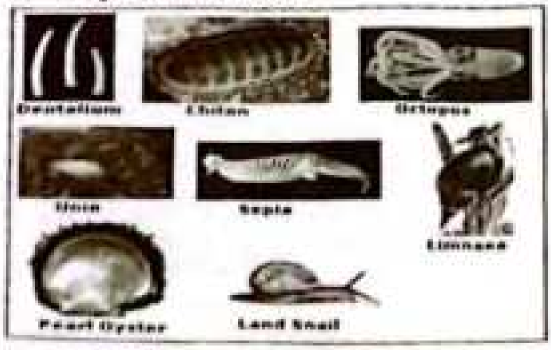


Figure 19: Phylum Mollusca

Characteristics of Phylum Mollusca

- Level of Organization – Organ systems, The cells have three layers– called **Triploblastic**
- Symmetry – Bilaterally symmetrical
- Segmentation – Little segmentation
- Body Cavity/ Coelom – Reduced
- Presence of Organs – Definite organs
- they are divided into three

- Examples – Snails
- Body is divided into head, Visceral Mass and Muscular Foot.
- Some of the molluscs have hard external shell like that of Snails and some have internal reduced shell like that in Octopus.
- They have an open circulatory system
- There is a kidney-like organ for excretion

8. Phylum Echinodermata

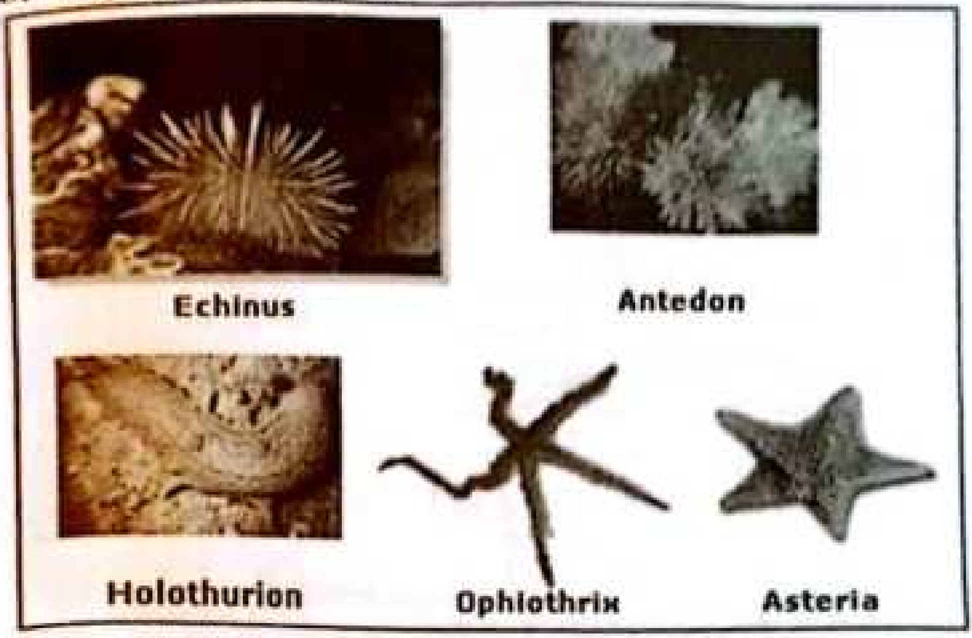


Figure 20: Phylum Echinodermata

Characteristics of Phylum Echinodermata

- Level of Organization – Organ systems, the cells have three layers — called **Triploblastic**
- Symmetry – Bilaterally symmetrical in larval stage and Radially symmetrical in Adults.
- Segmentation – No
- Body Cavity/ Coelom – True body cavity
- Presence of Organs – Definite organs
- Examples – Starfish, Sea cucumber
- They have Spiny dermis made of calcium carbonate
- They have a water vascular system which helps in feeding and locomotion.

9. Phylum Chordata

Characteristics of Phylum Chordata-

- They have a notochord. It is a rod-shaped structure that provides skeletal support to body. It is found in the embryonic stage of all chordates and in adult stages for so chordates.
- A nerve cord that connects brain.

- Most aquatic animals have a Pharyngeal slit that allows the exit of water
- They have a post-anal Tail made up of muscles and skeletal elements that helps in balancing.

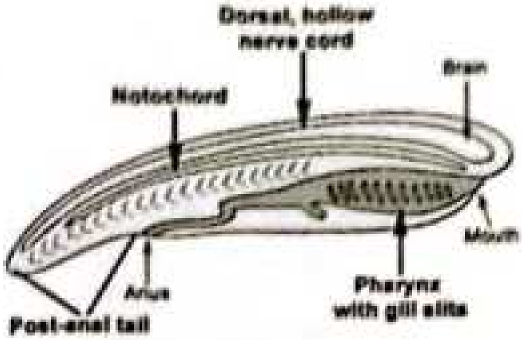
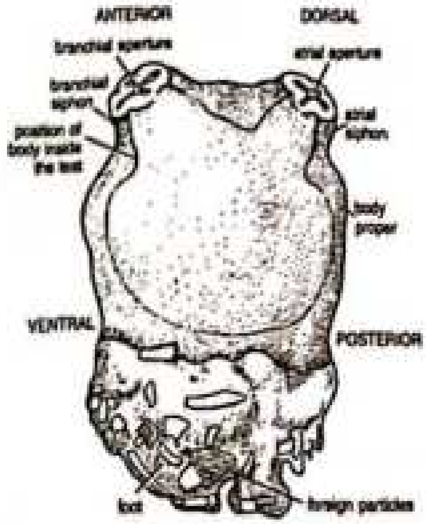


Figure 21: Phylum Chordata

Subphylum Protochordate



Herdmania

Figure 22: Subphylum Protochordate

Characteristics of Subphylum Protochordate

- Level of Organization – Organ systems, the cells have three layers– called **Triploblastic**
- Notochord present in some stage of life.
- Symmetry – Bilaterally symmetrical
- Segmentation – No
- Body Cavity/ Coelom – Present
- Presence of Organs – Definite organs
- Examples – Ascidia, Herdmania

CamScanner

Figure 23: Subphylum Vertebrata

Characteristics of Subphylum Vertebrata

- Level of Organization – Organ systems, highly developed tissues, the cells have three layers – Upper layer and the inner layer – called **Triploblastic**
- Symmetry – Bilaterally symmetrical
- Segmentation – Yes
- Body Cavity/ Coelom – Present, well-defined
- Presence of Organs – Definite organs
- Examples – Mammals , Birds, Fishes
- They have vertebral column developed from notochord.
- The internal skeleton muscles can attach at various points of the body
- There is a dorsal hollow nerve cord in the upper side of the back
-

Cold-blooded Animals and Warm- blooded Animals

Cold-blooded Animals	Warm-blooded Animals
They cannot maintain a constant body temperature	They can maintain a constant body temperature
They obtain heat from the environment surrounding them	They obtain heat from the food they eat

Their body temperature can vary as per the surrounding temperature	They maintain a temperature of around 35 – 40 degree Celsius irrespective of the surrounding temperature
They regulate heat in their bodies by changing colours or by being in sunlight	They regulate their body heat by metabolic processes and adaptive mechanisms such as hibernation and sweating.
Examples – Fishes, Reptiles, Insects, Amphibians	Examples – Mammals and Birds

	Body Type	Heart Chambers	Cold-blooded / Warm-blooded (Body Temperature)	Respiration	Reproduction	Found at	Examples
Pisces / Fish	They have scales or plates on their body, a muscular tail, some have skeleton made up of cartilage, some have skeleton made up of bones and cartilage	2 chambers	cold blooded	gills	Eggs	Water	Synchiropus splendidus (Mandarin fish), Scoliodon (Dog fish)
Amphibian	Have smooth and slimy skin	3 Chambers	cold blooded	Gills in larval stage and lungs in adult stage	Eggs	Land and water.	Toad, Hyla (Tree frog)
Reptilia	Have dry scales	3 Chambers except Crocodile which has 4 heart chambers	Cold blooded	lungs	Eggs	Land, Water	Turtles, King Cobra
Aves / Birds	They have waterproof skin which is covered with feathers. They have a beak or bill rather than teeth. Their forelimbs are developed into wings. They have hollow bones or pneumatic bones	4 Chambers	Warm blooded	Lungs	Eggs	Land, air	Crow, Pigeon

[illegible]	Have skin with hair and sweat glands	4 Chambers	Warm blooded	Lungs	give birth to young ones except Duck billed Platypus and Echidna	Land, water, air	Humans Cats

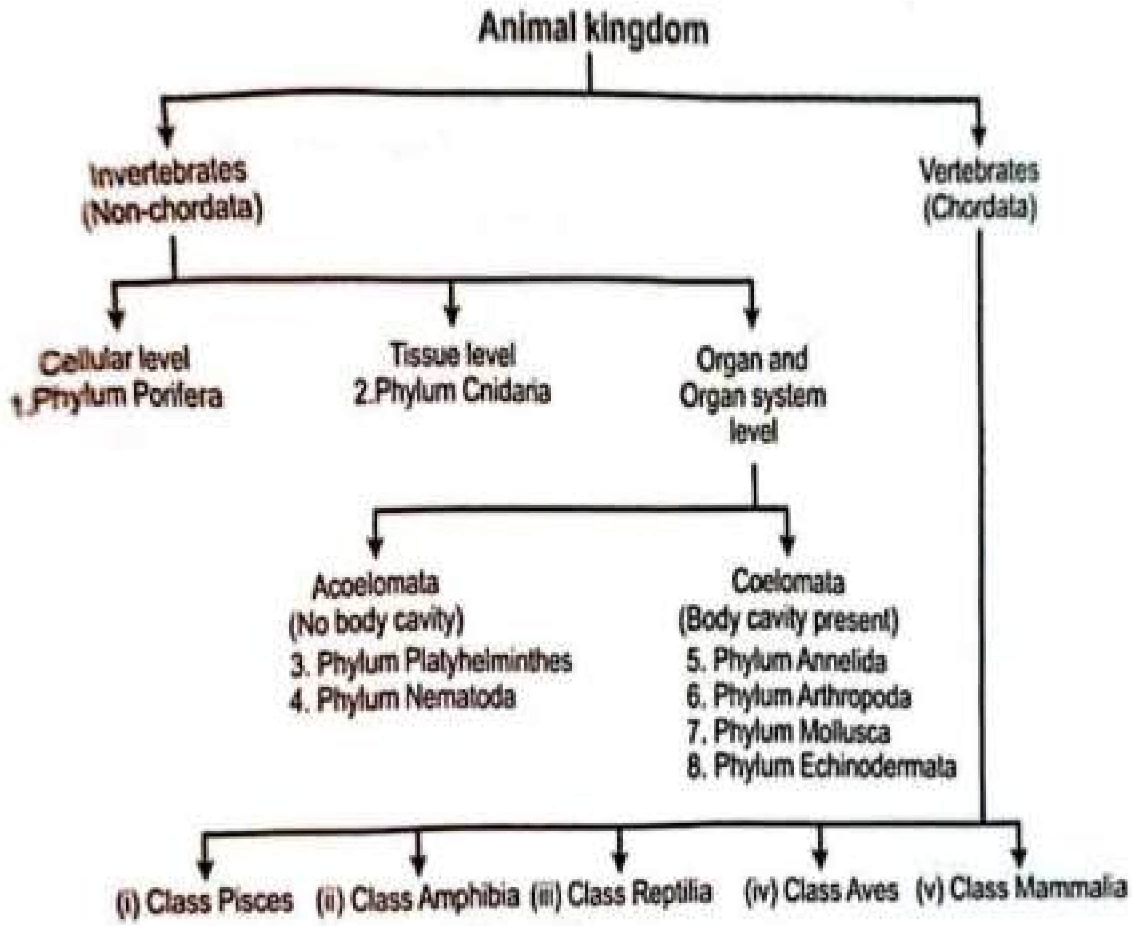


Figure 24: Classification of Animalia Kingdom

CHAPTER 3

Cytology

Cytology is the study of cells. There are two types: prokaryotic cells, the structure of which i described, and eukaryotic cells.

Table 1.0: Comparisons of Prokaryotic and Eukaryotic Cells

Prokaryotic	Eukaryotic cells
No distinct nuclear; only diffuse area(s) nucleoplasm with no nuclear membrane	A distinct, membrane-bounded nucleus
No chromosomes-circular strands of DNA	Chromosomes present on which DNA located
No membrane-bounded organelles organelles such as chloroplasts and mitochondria	Chloroplasts and mitochondria may present
Ribosomes are smaller	Ribosomes are larger
Flagellant (if present) lack internal 9+2 fibril arrangement	Flagella have 9 + 2 internal f arrangement
No mitosis or metals occurs	Mitosis and/or meiosis occurs

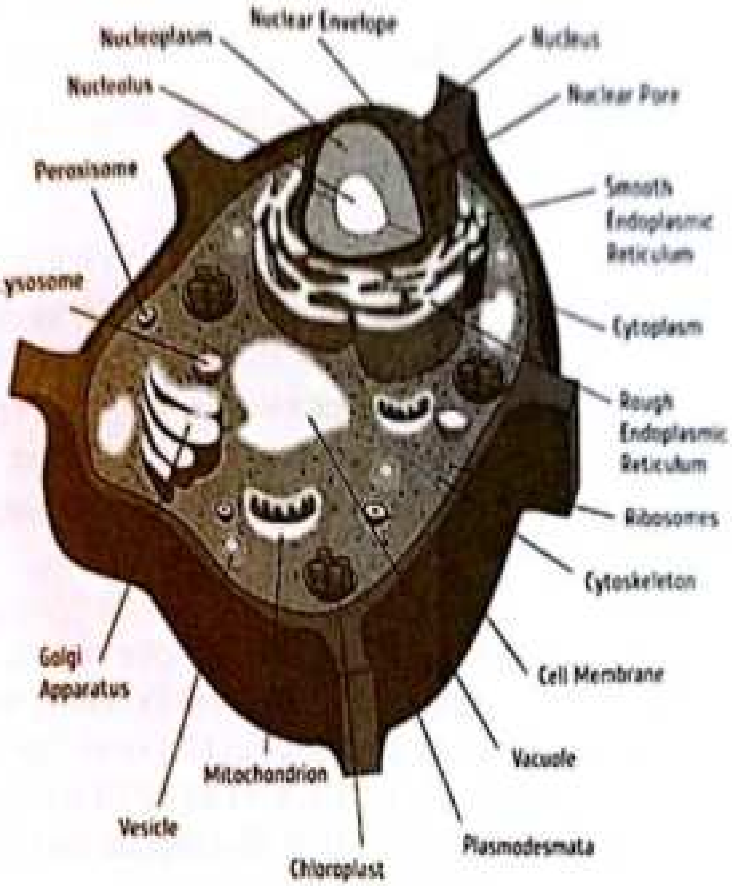
Structure of Eukaryotic Cells

The vast majority of organisms are composed of eukaryotic cells. They are of two basic plant cells and animal cells.

Table 1.1: Plant and Animal Cell Differences

Plant	Animal
Cellulose cell wall as well as membrane	No cellulose cell wall, only membrane
Pits in cell wall	No pits
Plasmodesmata present	No plasmodesmata
Large vacuole filled with cell sap	Some vacuoles but usually small and numerous
Cytoplasm peripheral	Cytoplasm throughout the cell
Nucleus usually peripheral	Nucleus anywhere in cytoplasm but central

Two cytoplasmic membranes, outer plasmalemma, Inner tonoplast	Only one cytoplasmic membrane
Variety of plastids, e.g. chloroplast, leucoplast	Not normally any plastids
Cilia and flagella absent in higher plant	Cilia commons in higher animals
Centrioles absent in higher plants	Centrioles present



Plant Cell

Cell Organelles

To study cell organelles in detail, the different parts of the cell are often separated- a process called **cell fractionation.** Cells are broken up and placed in a **centrifuge,** an instrument which spins at high speed to exert a force causes the different parts to settle out at different spin-speeds **(differential centrifugation)**

Cell Membrane

The cell membrane acts as a boundary between a cell and its environment. It permits different substance to pass through it at different rates and is therefore described as being **partially permeable.** All cell membranes, whether they surround cells or organelles, are basically the same and are known as **unit membranes**. The generally accepted structure is the **fluid mosaic model** (see Fig.1.10) proposed by J Singer and GL Nicholson in 1972.

Nucleus

If you have ever viewed a eukaryotic cell under a microscope you will probably have noticed the prominent nucleus. It is bounded by a double unit membrane, the nuclear envelope, which contains pores 40-100 nm in diameter. The outer membrane is granular whereas the inner one is smooth. Within the membrane is a mesh of nucleoplasm interspersed with nuclear ribosomes and chromatin (DNA bound to protein). There may also be one or two small, round bodies - the **nucleoli**, which store RNA. The functions of the nucleus are to

1. act as the control centre for the cell;
2. contain the genetic material of the cells;
3. produce ribosomes and RNA;
4. play an essential role in cell division.

Mitochondria

These vary in shape and size but are usually rod shaped, typically being 5µm in length and 0.2µm in diameter. The wall consists of two thin membranes, the inner one of which is folded to form **cristae.** Cell respiration takes place in mitochondria, most of the necessary enzymes being attached to the inner membrane and cristae, or occurring in the matrix. Thus cells which expand a lot energy have numerous mitochondria, each with many cristae, e.g. muscle cells.

Endoplasmic Reticulum

If you study an electron micrograph of a cell you will readily observe in the cytoplasm a network of membranes. This is the **endoplasmic reticulum (ER),** which forms a cytoplasmic skeleton. An extension of the outer nuclear membrane, the ER forms a series of sheets which enclose flattened sacs called **cisternae** (see fig. 1. 13). Where the ER is lined with ribosomes it is known as **rough endoplasmic reticulum**; where ribosomes are absent the term **smooth endoplasmic reticulum** is used. The functions of ER are to:

1. form a transport network throughout the cell;
2. provide a large surface area for chemical reactions;
3. play a role in protein synthesis (rough ER);
4. collect and store manufactured material;
5. form a structural skeleton to help maintain the shape of the cell;
6. produce lipids and steroids (smooth ER)

Golgi apparatus (dictyosome)

Similar in structure to smooth ER, the Golgi apparatus is composed of membrane-bound flattened sacs piled up in stacks. The Golgi apparatus is especially well developed in secretory cells. Its functions include:

- production of glycoproteins;
- secretion of carbohydrates involved in the production of new cell walls;
- production and secretory enzymes;

4. transport and storage of lipids;
5. formation of lysosomes.

Lysosomes

These are spherical organelles 0.1- 1.0μm in diameter, which contain around 50 enzymes, mostly hydrolases, in acid solution. Their functions include;

1. the destruction of unwanted or worn-out cell organelles;
2. the digestion of material engulfed by a cell, e.g. bacteria engulfed by white blood cells;
3. the release of enzymes outside the cell **(exocytosis)** to digest external material;
4. the complete digestion of a cell after its death **(autolysis)**

Ribosomes

These are small cytoplasmic granules about 20nm in diameter in eukaryotic cells (80S type) but slightly smaller in prokaryotic cells (70S type). In groups they are known as **polysomes.** Ribosomes are made up of small RNA molecules and protein and are important in protein synthesis (see3.1).

Peroxisomes (microbodies)

These are small, spherical, membrane-bound bodies 0.5- 1.5μm in diameter. They contain enzymes, in particular catalase which breaks down the highly toxic cellular by-product hydrogen peroxide into water and oxygen, thereby preventing the cell being poisoned. Peroxisomes are found in large numbers in actively metabolizing cells, such as liver cells.

Vacuoles

This is a general term for any fluid-filled sac surrounded by a membrane. In plant cells there is normally a single, large central vacuole surrounded by a membrane called a tonoplast. In plant cells this vacuole stores amino acids and sugars as well as certain waste, it contains coloured pigments and, most importantly, supports the herbaceous parts of a plant by providing an osmotic system which creates a pressures potential.

Storage Granules

These include starch grains within plant cells, glycogen granule in animal cells and oil or lipi droplets in both. All acts as energy stores.

Centrioles

In animal cells two centrioles are found at right angle to each other. They are hollow cylind about 0.2 Um in diameter and have the same basic structure as the basal bodies of cilia. Tl function in the formation of the spindle at cell division (see 3.2) and also give rise to cilia flagella.

Cilia and Flagella

Both may be concerned with locomotion in a cellular organism but cilia are also widespread in higher animal groups where they move material within an organism, e.g. cilia move mucus in the respiratory tract of mammals. Cilia are up to 25 um long and flagella 1 000 um long, both have an average diameter of less than 0.3 um. They have the same basic structure. At the base of each cilium is a basal body composed of nine peripheral fibres but not the central two. Movement of cilia and flagella requires ATP, and the bending process is initiated is initiated at the basic and transmitted towards the tip.

Microtubules

These slender, unbranched tubes around 24 nm in diameter are widespread in eukaryotic cells. Their functions include:

1. Providing an internal cellular skeleton (cytoskeleton);
2. Providing routes in cells along with materials move;
3. Forming a framework on which the cellular cell wall of plant cells laid down;
4. Aiding cell division by forming the spindle;
5. A major component of cilia and flagella.

Microfilaments

These very thin strands about 6mm in diameter are made up of the protein actin and sometimes myosin. They are probably involved in movement within cells.

Microvilli

Around 0.6um in length, the microvilli are tiny, finger-like projections of the cell membrane of certain cells such as those in the kidney tubule and the epithelium of the intestine. Collectively they form the brush border of these cells and help to increase their surface area aiding absorption.

Cellulose cell wall

You will probably have observed that plant cells, when seen through a microscope, have a distinct boundary layer. This is the cell wall- a characteristics feature of plant cells. It made of fibrils of the polysaccharide cellulose, which is often impregnated by other polysaccharides such as pectin and lignin. The function of the cell wall is to:

1. Provide support in herbaceous plants;
2. Allow movement of water through and along it;
3. Act as waterproofing layer when impregnated with other substances such as lignin.

Chloroplasts

These are found only in plant cells. Their shape and size vary with species but there is always a double until membrane surrounding a matrix (stroma) in which are stacks of lamellae (grana). The lamella holds the chlorophyll in the most suitable position for photosynthesis. The stroma contains numerous starch granules and enzymes for the reduction of carbon dioxide.

Transport of Materials into and Out of Cells

The numerous of am, trials in and out of cells is achieved by diffusion, osmosis, active transport, phagocytosis and pinocytosis

Diffusion

This is the process by which the particles of substance move from a region where it is highly concentrated to a region where its concentration is lower. Generally a slow process, the rate of diffusion is faster if:

1. The difference in concentration between the two regions (concentration gradient) is made greater;
2. The distance between the two regions ids decreased;
3. The area over which diffusion occurs is increased;
4. The molecules diffusion is small and fat-soluble.

CHAPTER 4

Reproduction

PLANT REPRODUCTIVE SYSTEM

Reproduction in plants is either asexual or sexual. Asexual reproduction in plants involves a variety of widely disparate methods for producing new plants identical in every respect to the parent. Sexual reproduction, on the other hand, depends on a complex series of basic cellular events, involving chromosomes and their genes, that take place within an elaborate sexual apparatus evolved precisely for the development of new plants in some respects different from the two parents that played a role in their production.

General features of asexual systems

Asexual reproduction involves no union of cells or nuclei of cells and, therefore, no mingling of genetic traits, since the nucleus contains the genetic material (chromosomes) of the cell.

Reproduction by fragments

In many plant groups, fragmentation of the plant body, followed by regeneration and development of the fragments into whole new organisms, serves as a reproductive system. Fragments of the plant bodies of liverworts and mosses regenerate to form new plants. In nature and in laboratory and greenhouse cultures, liverworts fragment as a result of growth; the growing fragments separate by decay at the region of attachment to the parent. During prolonged drought, the mature portions of liverworts often die, but their tips resume growth and produce a series of new plants from the original parent plant.

In mosses, small fragments of the stems and leaves (even single cells of the latter) can, with sufficient moisture and under proper conditions, regenerate and ultimately develop into new plants.

Reproduction by special asexual structures

Throughout the plant kingdom, specially differentiated or modified cells, groups of cells, or organs have, during the course of evolution, come to function as organs of asexual reproduction. These structures are asexual in that the individual reproductive agent develops into a new individual without the union of sex cells (gametes).

Airborne spores characterize most nonflowering land plants, such as mosses, liverworts, and ferns. Although the spores arise as products of meiosis, a cellular event in which the number of chromosomes in the nucleus is halved, such spores are asexual in the sense that they may grow directly into new individuals, without prior sexual union.

The vegetative, or somatic, organs of plants may, in their entirety, be modified to serve as organs of reproduction. In this category belong such flowering-plant structures as stolons, rhizomes, tubers, corms, and bulbs, as well as the tubers of liverworts, ferns, and horsetails, the dormant buds of certain moss stages, and the leaves of many succulents. Stolons are elongated runners, or horizontal stems, such as those of the strawberry, which root and form new plantlets when they make proper contact with a moist soil surface. Rhizomes, as seen in ginger, are fleshy, elongated, horizontal stems that grow within or upon the soil. The branching of rhizomes results in multiplication of the plant. The enlarged fleshy tips of subterranean rhizomes or stolons are known as tubers, examples of which are potatoes. Tubers are fleshy storage stems, the buds ("eyes") of which, under proper conditions, can develop into new individuals. Erect, vertical, fleshy, subterranean stems, which are known as corms, are exemplified by sugarcane. These organs tide the plants over periods of dormancy and may develop secondary cormlets, which give rise to new plantlets. Unlike the corm, only a small portion of the bulb, as in the onion, represents stem tissue. The latter is surrounded by the fleshy food-storage bases of earlier-formed leaves. After a period of dormancy, bulbs develop into new individuals. Large bulbs produce secondary bulbs through development of buds, resulting in an increase in number of individuals.

General features of sexual systems

In most plant groups, both sexual and asexual methods of reproduction occur.

The cellular basis

Sexual reproduction at the cellular level generally involves the following phenomena: the union of sex cells and their nuclei, with concomitant association of their chromosomes, which contain the genes, and the nuclear division called meiosis. The sex cells are called gametes and the product of their union is a zygote. Gametes may be motile, by means of whiplike hair (flagella) or of flowing cytoplasm (amoeboid motion). In their union, gametes may b morphologically indistinguishable (i.e., isogamous) or they may be distinguishable only on tl criterion of size (i.e., heterogamous). The larger gamete, or egg, is nonmotile; the smal gamete, or sperm, is motile. In oogamous reproduction, the union of sperm and egg called fertilization.

In bryophytes and vascular plants, sexual reproduction is of the oogamous type, modification thereof, in which the sex cells, or gametes, are of two types, a larger nonm egg and a smaller motile sperm. These gametes are often produced in special cont called gametangia, which are multicellular. In oogamy, the male gametangi called antheridia and the female oogonia or archegonia. A female gametangium with a cellular jacket is called an archegonium, and produces eggs. In most plants, the e produced in archegonia and the sperms in antheridia with surface layers of sterile cell

The plant basis

Individual plants may be either bisexual (hermaphroditic), in which male and female gametes are produced by the same organism, or unisexual, producing either male or female gametes but not both. A bisexual individual, however, is not necessarily capable of fertilizing its own eggs.

Among the liverworts, mosses, and vascular plants, the life cycle involves two different phases, often called generations, although only one plant generation is, in fact, involved in a complete cycle. This type of life cycle is often said to illustrate the "alternation of generations," in which a haploid individual (i.e., with one set of chromosomes), or tissue, called a gametophyte, at maturity produces gametes that unite in pairs to form diploid (i.e., containing two sets of chromosomes) zygotes. The latter develop directly into individuals, or tissues, called sporophytes, in which the nuclei of certain fertile cells, called spore mother cells, or sporocytes, give rise to haploid spores (sometimes called meiospores). These spores are lightweight and are borne by air currents; they germinate to form the haploid, sexual, gamete-producing phase, usually designated the gametophyte.

Bryophyte reproductive systems - **Liverworts and hornworts**

The plant bodies of liverworts and hornworts represent the gametophytic (sexual) phase of the life cycle, which is dominant in these plants. In the liverworts, the sporophyte is borne upon or within the gametophyte but is transitory. Liverwort and hornwort plants, depending on the species, may be bisexual or unisexual, and the sex organs may be distributed on the surface (*Riccia, Ricciocarpus, Sphaerocarpos,Pellia*) or localized in groups and borne on special branches (antheridiophores and archegoniophores), as in *Marchantia*; the sperm cells are biflagellate.

Release of the mature sperm and the process of fertilization require moisture in the form of heavy dew or raindrops. In all but a few genera (*Riccia, Ricciocarpus*), the developing sporophytes are actively photosynthetic—i.e., capable of utilizing light energy to form organic substances.

Mosses

In mosses, as in liverworts and hornworts, the leafy shoots belong to the gametophytic phase and produce sex organs when they mature. The leafy shoots (often called gametophores because they bear the sex organs) arise from a preliminary phase called the protonema, the direct product of spore germination. Filamentous, straplike, or membranous, it grows along the soil surface. A protonema of a moss may proliferate, apparently indefinitely, under favourable conditions and thus increase the population of leafy shoots that arise as buds. Under adverse conditions, certain buds and branches of the protonema may thicken their walls and serve to tide the species over an unfavourable growing period.

Antheridia and archegonia may be borne at the tips (apices) of the main shoots or on special lateral branchlets. Both bisexual and unisexual leafy shoots occur, depending on the species. In a number of mosses (*Mnium, Polytrichum, Funaria*), the sexually mature shoots become recognizable through the production of special prominent leaves that form an apical cup around the sex organs. If the cup is brightly coloured, it is often flowerlike.

The archegonia and antheridia of mosses are large enough in many species to be just visible to the unaided eye. The jacket cells of the antheridia are often coloured bright orange or rust; their sperm are biflagellate. As in liverworts and hornworts, rains and even heavy dews evoke the liberation of sperm and the opening of the mature archegonia so that fertilization may be accomplished.

The moss sporophyte, which is attached to the gametophyte, photosynthesizes during much of its development and is more or less self-supporting.

After elongation of the moss sporophyte has ceased, the distal portion (farthest away) enlarges to form the capsule (sporangium), or spore-bearing region. The spores (meiospores), which arise by meiosis, are shed from the capsule gradually through a variety of mechanisms. After the operculum (cover) of the capsule has been shed, its mouth is usually partially closed by the peristome (teeth) and sometimes by associated structures. These teeth absorb moisture, and their resultant swelling and contraction open spaces through which the spores are shed.

TRACHEOPHYTE (VASCULAR CRYPTOGAMS AND SEED PLANTS) REPRODUCTIVE SYSTEMS

Spore plants

In liverworts, hornworts, and mosses, the dominant phase in the life cycle is the sexual gametophyte. In the tracheophytes (vascular cryptogams [which lack true flowers or seeds] and seed plants), on the other hand, the sporophyte is the dominant phase in the life cycle. The gametophytes of the vascular cryptogams mature after the spores that initiated them have been shed from the parent plant, so the gametophytes are free-living. In the seed plants, the gametophytes mature as parasites on the sporophytes.

PSILOPSIDS

The trilobed sporangia of the whisk fern *Psilotum*—not a true fern but a psilopsid—are borne terminally on short lateral branches. During development, some of the potentially spore-bearing tissue is used as nutrient by the sporocytes as they complete the meiotic divisions that result in colourless kidney-shaped spores. The spores, which are shed as the sporangia open along three lines, germinate and slowly develop into cylindrical sparingly branched gametophytes. They presumably derive nourishment from decaying matter and occur in humus-rich soil, in rock fissures, or among roots on the trunks of tree ferns. The cells of the gametophyte contain fungal structures (hyphae) that probably are involved in some type of nutritional relation with the gametophyte.

The gametophytes are bisexual and the sperm cells are multiflagellate. The embryonic ...hyte is not easily distinguished from the gametophyte that bears it. At first anchored to ...hyte by an absorptive foot, the sporophyte ultimately becomes separated from being ... the gametophyte.

...IDS

In the genus *Lycopodium*, the sporangia are closely associated with the leaves. In some species (*L. lucidulum*), the sporangium-bearing leaves (sporophylls) occur in zones among the vegetative portions of the stems. In most, however, the sporophylls occur in specialized compressed stems called cones or strobili.

In *Selaginella* (Spike mosses), the sporophytes have sporophylls localized in strobili. All species of *Selaginella* are heterosporous i.e. produces spores of two sizes; the larger designated as megaspores which develop into female gametophyte and the smaller known as microspores which become the male gametophyte. The spores of *Selaginella* begin to develop into gametophytes before they have been shed from their sporangia and attain maturity on a suitable moist substrate.

The microscopic male gametophyte is composed essentially of a single antheridium, which produces biflagellate sperm. The female gametophyte consists of vegetative cells with several archegonia at maturity. Both male and female gametophytes lack chlorophyll (green pigment), which is necessary for photosynthesis, thus they utilize nutrients stored in the spores.

FERNS

As they mature, many fern sporophytes begin to produce spores in clusters of sporangia on the undersurfaces of their fronds (or vegetative "leaves"). Others produce their sporangia on highly modified leaves or portions thereof.

The site of origin of the sporangia is the receptacle, this with its groups of sporangia, is called a sorus. In many ferns, each sorus is covered with a special outgrowth, the indusium; in others the sporangia are covered during development by the margin of the leaf. In a few ferns (e.g., *Polypodium*), the sori remain uncovered.

Seed plants

In the two great groups of seed plants, gymnosperms and angiosperms, the sporophyte is the dominant phase in the life cycle, as it is also in the vascular cryptogams; the gametophytes are microscopic parasites on the sporophytes.

n the gymnosperms, the seeds occur individually, exposed at the ends of stalks, sometimes in horls on an axis, or on the scales of a cone, or megastrobilus. In angiosperms, flowering plants, on the other hand, the seeds are enclosed during development in a structure riously termed a pistil or a carpel, which is sometimes considered to represent an folded megasporophyll.

A number of parts of the reproductive process are common to both angiosperms and gymnosperms: (1) they produce seeds at maturity; (2) the megasporangium, unlike that of heterosporous seedless plants, is covered by one or two cellular layers called integuments and is termed an ovule; (3) there is a minute passageway, or micropyle, through the integuments; (4) the ovule matures as a seed; (5) only one megasporocyte is present and undergoes meiosis in the megasporangium to produce four megaspores, only one of which usually is functional; (6) the megaspore is never discharged from its megasporangium and ovule; (7) one female gametophyte is produced within each megasporangium and ovule; (8) the microspores begin their development into male gametophytes while still enclosed in the microsporangia; (9) as they mature, the male gametophytes, which are contained within the microspore wall and are termed pollen grains, develop a tube that conveys sperm to the egg cell; (10) union of sperm and egg and development of an embryonic sporophyte from the zygote occur within the female gametophyte (sometimes called the "embryo sac"), which is covered by the remains of the megasporangium and by integuments; (11) as the embryo develops, the ovule matures as a seed.

Differences, which, in addition to seed position, serve to distinguish angiosperms from gymnosperms. (1) The reproductive cycle in most angiosperms is completed more quickly than that in gymnosperms, and the gametophytes are smaller and simpler and, (2) unlike those of most gymnosperms, lack archegonia. (3) The pollen in angiosperms is transferred to the surface of the megasporophyll, whereas in gymnosperms it is brought to the micropyle of the ovule itself. (4) Two sperm cells are involved in the sexual union in angiosperms: one unites with the egg to form a zygote; the other unites with two nuclei of the female gametophyte to form the primary endosperm nucleus. (5) This divides to form a post fertilization storage tissue, which serves as a food source for the embryo; the embryo of gymnosperms is nourished by the somatic (non-reproductive) tissues of the female gametophyte. (6) The angiosperm ovule increases to mature seed size after fertilization, whereas in gymnosperms, this enlargement occurs prior to fertilization.

Reproduction in Animals

Modes of Reproduction

There are two modes by which animals reproduce. These are: (i) Sexual reproduction, and (ii) Asexual reproduction.

Sexual Reproduction

In animals, males and females have different reproductive parts or organs. Like plants, the reproductive parts in animals also produce gametes that fuse to form a zygote. It is the zygote which develops into a new individual. This type of reproduction beginning from the fusion of male and female gametes is called sexual reproduction.

Male Reproductive Organs

The male reproductive organs include a pair of testes (singular, testis), two sperm ducts and a penis. The testis produces the male gametes called sperms. Millions of sperms are produced by the testes. Though sperms are very small in size, each has a head, a middle piece and a tail. Each sperm is a single cell with all the usual cell components.

Female Reproductive Organs:

The female reproductive organs are a pair of ovaries, oviducts (fallopian tubes) and the uterus. Ovary produces female gametes called ova (eggs). In human beings, a single matured egg is released into the oviduct by one of the ovaries every month. Uterus is the part where development of the baby takes place. Like the sperm, an egg is also a single cell.

Fertilization:

The first step in the process of reproduction is the fusion of a sperm and an ovum. When sperms come in contact with an egg, one of the sperms may fuse with the egg. Such fusion of the egg and the sperm is called fertilization. During fertilization, the nuclei of the sperm and the egg fuse to form a single nucleus. This results in the formation of a fertilized egg or zygote.

The process of fertilization is the meeting of an egg cell from the mother and a sperm cell from the father. So, the new individual inherits some characteristics from the mother and some from the father.

Fertilization which takes place inside the female body is called internal fertilization. Internal fertilization occurs in many animals including humans, cows, dogs and hens.
In many animals fertilization takes place outside the body of the female. In these animals, fertilization takes place in water.

During spring or rainy season, frogs and toads move to ponds and slow flowing streams. When the male and female come together in water, the female lays hundreds of eggs. Unlike hen's egg, frog's egg is not covered by a shell and it is comparatively very delicate. A layer of jelly holds the eggs together and provides protection to the eggs.

As the eggs are laid, the male deposits sperms over them. Each sperm swims randomly in water with the help of its long tail. The sperms come in contact with the eggs. This results in fertilization. This type of fertilization in which the fusion of a male and a female gamete takes place outside the body of the female is called external fertilization. It is very common in aquatic animals such as fish, starfish, etc.

Development of Embryo

Fertilization results in the formation of zygote which begins to develop into an embryo. The zygote divides repeatedly to give rise to a ball of cells. The cells then begin to form groups that develop into different tissues and organs of the body. This developing structure is termed an embryo. The embryo gets embedded in the wall of the uterus for further development.

The embryo continues to develop in the uterus. It gradually develops the body parts such as hands, legs, head, eyes, ears, etc. The stage of the embryo in which all the body parts can be identified is called a foetus. When the development of the foetus is complete, the mother gives birth to the baby.

In animals which undergo external fertilization, development of the embryo takes place outside the female body. The embryos continue to grow within their egg coverings. After the embryos develop, the eggs hatch.

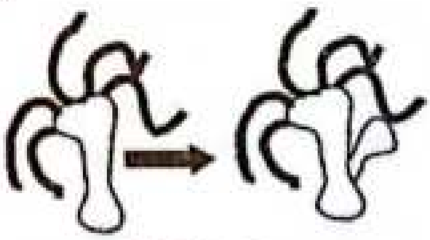
Viviparous and Oviparous Animals

Some animals give birth to young ones while some animals lay eggs which later develop into young ones. The animals which give birth to young ones are called viviparous animals. Those animals which lay eggs are called oviparous animals. Ovoviviparity (aplacental viviparity) is a mode of reproduction in sharks, fishes, amphibians (and other animals) in which embryos develop inside eggs that are retained within the mother's body until they are ready to hatch. Ovoviviparous animals are similar to viviparous species in that there is internal fertilization and the young are born live, but differ in that there is no placental connection and the unborn young are nourished by egg yolk. However, the mother's body does provide gas exchange (respiration).

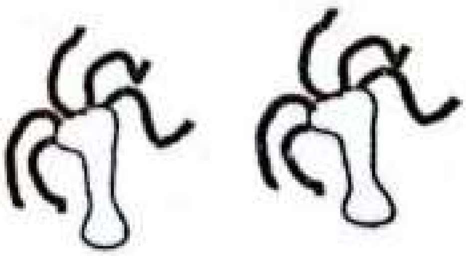
A Sexual Reproduction

The type of reproduction in which only a single parent is involved is called asexu reproduction.

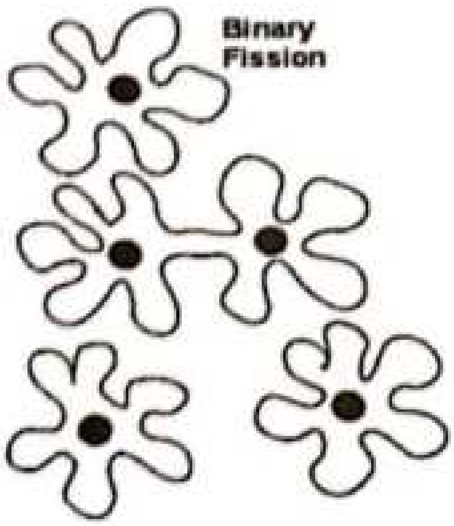
In each hydra, there may be one or more bulges. These bulges are the developing n individuals and they are called buds. In hydra too the new individuals develop as outgrov from a single parent. Since new individuals develop from the buds in hydra, this type of ase reproduction is called budding.



Budding



Another method of asexual reproduction is observed in the microscopic organism, amoeba. Amoeba is a single-celled organisms. It begins the process of reproduction by the division of its nucleus into two nuclei. This is followed by division of its body into two, each part receiving a nucleus. Finally, two amoebae are produced from one parent amoeba. This type of asexual reproduction in which an animal reproduces by dividing into two individuals is called binary fission.



CHAPTER 5

Interrelationship of Organism

some key relationships in the life of an organisms are its interactions with individuals of other species in the community. These interspecific interactions include competition, -predation, and symbiosis including parasitism.

Competition

This is an interaction that occurs when individuals of the same or different species fight or compete for a resource that limits their growth and survival. Animals could compete for mates, food, light, shelter or space. A shortage of a needed resource could increase the competition between organism, and some of the losers may even starve to death if the resource is food. For example, weeds growing in a garden compete with garden plants for soil nutrients and water. Grasshoppers and bison in the great plants compete for the grass they both eat. Both vegetable ..nts and the weeds need sunlight water and minerals in order to survive. Dogs and cats need ..me kind of food and may fight over it.

Predation

Predation refers to an interaction between species in which one species, the predator, kills and eats the other, the pray.

Example include: Lion attacking and eating antelope, the hawk catches the fish for food, frogs feed on insects, etc.

Many important feeding adaptations of predators are both obvious and familiar. Most predators have acute senses that enable them to locate and identify potential prey. In addition, many predators have adaptations such as claws, teeth, stingers or poisons that help them catch and subdue the organisms on which they feed. Predators that pursue their prey ate generally fast and agile, whereas those that lie in ambush often disguised in their environments.

Prey animals have adaptations that help them avoid being eaten. Some common behavioural defences are hiding, fleeing and forming herds or schools. Other behavioural defences include alarm classes that summons many individuals of the prey species, which then mob the predator

Symbiosis

When individuals of two or more species live in direct and intimate contact with one anoth.. ..eir relationship is called symbiosis. It includes all such interactions, whether harmful, help.. .. neutral.

Parasitism

Parasites is a symbiotic interaction in which one organism, the parasite, derives its nourishment from another organism, its host, which is harmed in the process. Parasites that live within the body of their host is called endoparasites e.g. tapeworms, hookworms or roundworms in human digestive tract, while parasites that feed on the external surface of a host is called ectoparasites e.g. ticks and lice.

Parasites can significantly affect the survival, reproduction and density of their host populations either directly or indirectly. For example, ticks that live as ectoparasites on moose weaken their hosts by withdrawing blood and causing hair breakage and loss, increasing the chance that the moose will die from cold stress or predation by wolves.

Mutualisms

This is a biological interacting between two organisms in which both organism benefits fr the association. Example include; the association between clown fish and sea Anemones.

1. The sea Anemones provide the clown fish with protection from predators while clown fi defends the Anemones from butterfly fish who like to eat Anemones.
2. A remarkable 3-way mutualism appears to have evolved between an ant, a butter caterpillar, and an acacia in the American Southwest. The caterpillars have nectar org which the ants drink from, the acacia tolerates the feeding caterpillars. The ants appear provide some protection for both plant and caterpillar.
3. Nitrogen Fixation by bacteria in the root nodules of legume
4. The digestion of cellulose by microorganism in the digestive systems of termites ruminants mammals.

The interaction between termites and the microorganism in their digestive system i example of <u>obligate mutualism</u>, in which at least one species has lost the ability to su without its partner. In <u>facultative mutualism</u>, both species can survive alone.

Mutualistic relationship sometimes involve the evolution of related adaptations in both sp with changes in either species likely, to affect the survival and reproduction of the othe example, most flowering plants have adaptations such as nectar or fruit that attract ar which function in pollination r seed dispersal. In turn, many animals have adaptations th: them find and consume nectar.

Commensalism

This is an interaction between species of organism in which one of the organisms ben the other is neither harmed nor helped. For example, the 'hitchhiking' species, such a hat live on the shells of aquatic turtles.

The hitchhikers gain a place to grown while having seeming by little or no effect on their ride.

Orchids and some kinds of fern are aerial plants. They usually grow on trunks or branches of trees. These plants get moisture and nutrients from the bank of the tree. They also use the tree for support because they do not have stems. They orchids or ferns do not seem to harm or help the tree.

The barnacles are shelled animals that cannot move on their own. They attached themselves to other animals like crabs or whales. The barnacles get transportation and a steady supply of food as the whale moves through the ocean. They whale on the other hand, is generally not affected by this kind of interaction.

CHAPTER 6

Heredity

Heredity is the transmission of characteristics or traits from parents to offspring.
It can also be defined as the inheritance of qualities or characters from one generation to the next.

N.B: A heritable feature that varies among individuals, such as flower colour, height is called a character. While each variant for a character, such as purple or white colour for flower, short or tall is called trait.

Variation

This is any difference between individual organism and groups of organism of any species caused by either genetic difference of by the effect of environmental factors. Variation can be shown in physical appearance metabolism, behaviour/learning and mental ability.

Types of Variation

Based on the types of cells, variation is classified into two types:

1. Somatic Variation
2. Genotypic Variation

Based on the degree of differences, variation is classified into two types:

1. Continuous Variation
2. Discontinuous Variation

Variation Based on Types of Cells

1. **Genotypic Variation:** These are variations caused by differences in the number or structure of chromosomes or b difference in the genes carried by the chromosome. Height, eye colour, body shape and some of the genotypic variations. Variation cannot be identified as genotypic by simply observing the organisms unless breeding experiments are performed under controlled environmental conditions.

2. **Somatic Variation:** These are variations that are caused by environmental factors, such as climate, food supply and activity of other organism. These variations are not due to differences in gene or chromosomes and in general are not transmitted to future generations. Hence, they are not significant in the process of evolution.

Variation Based on the Degree of Differences

1. Continuous Variation: This is variation that has no limit on the value that can occur within a population.
   a) These are fluctuating with environmental conditions.
   b) These are non-heritable
   c) They have no role in evolution
   d) They are most common and occur in all organisms.

   - Some examples of continuous variation are: Height, Weight, Heart rate, Finger length, and Leaf length.

2. **Discontinuous Variation**: This is variation that has distinct groups for organisms to belong to.
   a) They are relatively unaffected by environmental conditions.
   b) They are heritable
   c) They provide raw materials for evolution on which selection is based.
   d) They are not common and appear suddenly.

Some examples of discontinuous variation are: Tongue rolling, Fingerprint, Eye colour, Blood groups.

Mendel's Laws of Heredity/Inheritance

1. **Law of Dominance**: It states that when two pure plants with contrasting characters are crossed, only one form of the character appears in F1 generation, the other remains unexpressed. The character which appears itself in F1 generation is called dominant, the alternative factor that fails to show itself in F1 generation is called recessive.

2. **Law of Segregation**: This states that the two alleles for a heritable character segregate (separate) during gamete formation and end up in different gametes.

3. **Law of Independent Assortment**: This states that each pair of alleles segregates independently of each other pair of alleles during gamete formation.

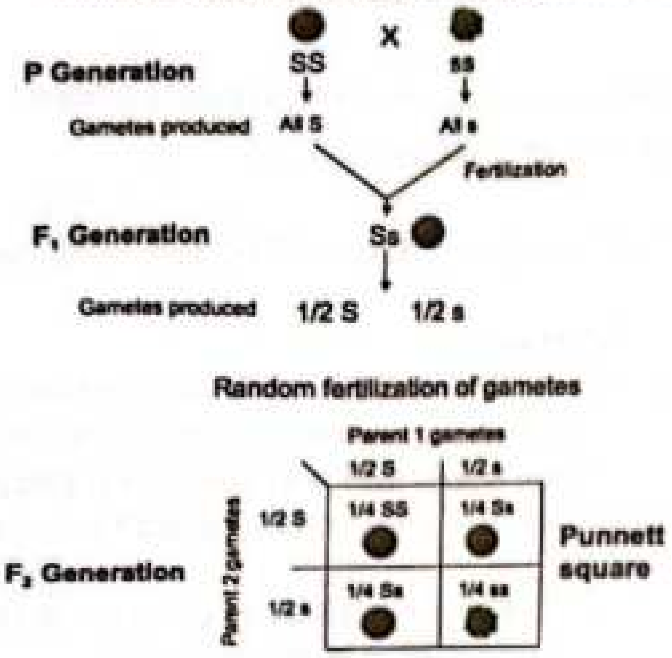
Mendel's monohybrid cross using symbols

P Generation	SS (Smooth)	X	ss (Wrinkled)
F_1 Generation		Ss	All smooth seeds
F1 X F1 cross	Ss	X	Ss
F_2 Generation	1/4 SS	1/2 Ss	1/4 ss

$\underbrace{\text{1/4 SS} \quad \text{1/2 Ss}}_{\text{Smooth seeds}} \quad \underbrace{\text{1/4 ss}}_{\text{Wrinkled seeds}}$

3 : 1 ratio

Mendel's monohybrid cross using Punnett square



- Note that in F_2 three types of genotypes are produced in 1:2:1 ratio, but because S is dominant, heterozygote have smooth phenotype, so phenotypic ratio is 3:1 smooth to wrinkled.
- Mendel analysed the behaviour of six other pairs of traits. All gave the same results:
  - results of reciprocal crosses were always the same.
  - all F_1 progeny resembled one of the parents, indicating dominance of one allele over the other.
  - trait that disappeared in F_1, reappeared in F_2, and there was always a 3:1 phenotypic ratio in F_2.

Useful Genetic Terms

1. **Allele:** This is an alternative versions of a gene that can occur at a particular locus on homologous chromosomes.
2. **Gene:** These are the units of inheritance which are passed from parents to the offspring.
3. **Character:** This is a heritable feature that varies among individuals.
4. **Trait:** A variant for a character.
5. **Dominant Allele:** This is an allele whose presence in a heterozygous genotype results in a phenotype characteristic of the allele.
6. **Recessive Allele:** This is an allele that does not express itself in the heterozygous condition.
7. **Genotype:** This is the genetic makeup of an individual
8. **Phenotype:** This is an organism's appearance or observable traits
9. **Homozygous:** A diploid with identity alleles at a locus
10. **Heterozygous:** Having two different alleles for a gene
11. **Monohybrid Cross:** This is a cross between individuals heterozygous for a sir character.
12. **Dihybrid Cross:** This is a cross between individuals heterozygous for two characters. YYRR and yyrr.
13. **Test Cross:** This is cross between an individual with an unknown genotype homozygous recessive individual.
14. **Back Cross:** This is the cross of an individual with one of its parents or with an or with the same genotype as a parent.
15. **A Hybrid** can be defined as an individual which result from the crossing of individual differing at least in one set of the character.
16. **Locus**: the site of a gene on a chromosome.
17. **P generation**: parental generation; always true breeding.
18. **F_1**: first filial generation; progeny of parentals
19. **F_2**: progeny of intermating F_1.
20. **Reciprocal cross**: A pair of crosses between a male of one strain and a female and vice versa is called reciprocal cross.
21. **Punnett square:** It is a square board with vertical and horizontal columns possible results of various crosses.

Sex Determination in Human Being

Genes determining the sex in human beings are located on a special pair of chro are called **sex chromosomes**. Accordingly, the 23 pairs of human chrom divided into two groups:

Autosomes: These are 22 pairs of chromosomes carry genes for body characters and general physiology.

Sex chromosome: This is a pair of chromosome carries genes for the determination of the sex. These are represented by X and Y chromosomes.

X- chromosome contains genes that determine female sex, Y- chromosome contains genes for male sex.

In humans, sex determination is of XX and XY types.

Female is homogametic, producing only one type of gametes (ova), each containing one X- chromosome

Male is heterogametic, producing two types of sperms, half carrying X and other half Y chromosome.

Man(heterogametic) = 44A + XY

erms: (22A+X) and (22A+Y)

Woman(homogametic)= 44A+XX

: (22A+X) and (22A+X).

Sperm	Ova	
	22A + X	22A + X
+ X	44A + XX Girl	44A + XX Girl
+ Y	44A + XY Boy	44A + XY Boy

CHAPTER 7

Evolution

olution can be defined as the sequence of gradual changes which tale place in the primitive anisms over millions of years in which new species are produced. Since the evolution is of ng organisms, so it is called organic evolution. It may also be defined as the, development organisms from pre-existing, less differentiated organisms over the course of time.

idences of Evolution

ne of the important sources which provide evidences for organic evolution are:

Fossil record: Fossils provide evidence of evolution. The remains of dead animals or plants that lived in the remote past are known as fossils. Fossils show the evolutionary changes that have occurred over time in various groups of organisms. Thus, fossils provide the evidence that the present animals (and plants) have originated from the previously existing ones through the process of continuous evolution.

Homologous organs: This provides evidence for evolutions. If we look at the ways in which living organisms are made, we can often see quite striking similarities in their construction but they may have very different functions. The presence of homologous forelimbs in a frog, a bird and a man indicate that all these forelimbs have evolved from a common ancestral animal which had a basic design limb. In other words, it tells us that a frog, a bird and a man, all have evolved from a common ancestor. Thus, the presence of homologous organs in different animals provides evidence for evolution by telling us that they are derived from the same ancestor who had the "basic design" of the organ on which all the homologous organs are based.

. **Analogous organs:** Those organs which have different basic structure (or different basic design) but have similar appearance and perform similar functions are called analogous organs. For example, the wings of an insect and bird have different structures but the same function. The presence of analogous organs indicates that even organisms having organ with different structures perform similar functions for their survival under hostil environmental conditions. Thus, the presence of analogous organs in different anima provide evidence for evolution by telling us that though they are not derived from comm ancestors, they can still evolve to perform similar functions to survive, flourish and ke on evolving in the prevailing environmental.

4. **Vestigial organs/structures:** Those organs in an organism which are functionless and useless now (but used to function in its ancestors) are intestine and the nictitating membranes of the eye are vestigial organs in human beings. Though the appendix is a vestigial organ in humans but the appendix is still functioning among the herbivorous ruminant mammals like cow and buffalo. This indicates that human beings may have evolved from such mammals which had a functional appendix in them. Similarly, the nictitating membrane in human (which is present as a small fold of skin in the corner of the eye) is a vestigial organ but is still functioning in birds and provides protection to their eyes. This indicates that human beings may have evolved from those ancestors who had a working nictitating membrane in the eyes. Thus, the presence of vestigial organs is a great evidence for the evolution of human beings.

5. **Embryology**: This also provides evidence for evolution. A study of the development of the embryo of an animal is called embryology. A study of the development of the embryos of the different vertebrate animals show striking similarities in their structure. In fact, the embryos of the different vertebrate animals are so similar in their early stages of development that it is difficult to distinguish one from the other.

Natural Selection: Is therefore, a process in which individuals that have certain heritable characteristics survive and reproduce at a higher rate than other individuals. OR it is a process by which nature selects and consolidates those organisms which are more suitably adapted and possess favourable variations.

The fact that the early embryos of all vertebrate animals like fish, salamander, tortoise, chicks and human look alike indicates that all these animals have evolved from a common ancestor.

Darwin's Theory of Evolution by Natural Selection

Darwin's theory of evolution by natural selection can be summarized as given below:

1. All organisms produce much more off springs than their environment can support. For example, a house fly lays thousands of eggs; a fern plant produces millions of spores, etc.
2. Due to over population, there is struggle between the members of same species as well as from other species for food, water, air, and space. Only a few are able to get it and survive and the rest perish and get eliminated.
3. Variations exist within population. The offspring of the same parents also differ from one another and show variations.

The individual with favourable variations survive in the struggle. The organisms with unfavourable variations die out and so only the fittest individuals survive, reproduce and transmit their favourable characters to the next generation. This process of the survival of the fittest over many generations slowly increases the population of fit individuals due to the heritable favourable variations. In other words, nature selects the individuals that are well adapted to the environment and allows them to survive. At the same time, nature rejects those that are poorly developed; hence, natural selection is a weeding process by which only the fittest individuals are selected.

5. As the favourable variations of the fittest animals are inherited to the offspring, these variations when accumulated over a long time, lead to the origin of new species.

Lamarckian Evolution

The French biologist Lamarck proposed, in 1809 a hypothesis to account for the mechanism of evolution based on two conditions; the use and disuse of parts, and the inheritance of acquired characteristics.

Changes in the environment may lead to changed patterns of behaviour which can necessitate new or increased use (or disuse) of certain organs or structures. Extensive use would lead to increased size and/or efficiency while disuse would lead to degeneracy and atrophy. These traits acquired during the lifetime of the individual were believed to be heritable and thus transmitted to offspring.

According to Lamarck, the long neck and legs of the modern giraffe were the result of generations of short-necked and short-legged ancestors feeding on leaves at progressively higher levels of trees. The slightly longer necks and legs produced in each generation were passed on to the subsequent generation, until the size of the present day giraffe was reached.

However, Lamarck's emphasis on the role of the environment in producing phenotypic changes in the individual was correct. For example, body-building exercises will increase the size of muscles, but these acquired traits, while affecting the phenotype, are non-genetic, and having on influence on the genotype cannot be inherited.

CHAPTER 8

Ecology

Element of Ecology and Type of Habitat

The word ***ecology*** is derived from the Greek root ***oikos***, which means "household," and ecology is indeed a study of the household of life. It is a ***holistic*** (broad-based and integrative) approach to understanding living things ***in context*** as they relate both to their physical environment (***abiotic*** aspects) and to each other (***biotic*** aspects). It is these ***interactions*** of living things that provide the raw material for ecological studies. It can also be defined as the study of the structure and the function of nature (Odum, 1963)

Basic Ecological Concept

A. Ecosystem

The concept of ecosystem was first put forth by Tansley (1935). Ecosystem is the major ecological unit. It has both structure and functions. The structure is related to species diversity. The more complex is the structure the greater is the diversity of the species in the ecosystem. The functioning of the ecosystem is related to the flow of energy and cycling of materials through structural component of the ecosystem. According to Odum, ecosystem is a basic functional unit of organisms and their environment interacting with each other and with their own components.

Factor affecting Ecosystem

Ecosystem consist of the following basic component

a. **Abiotic component** (non-living phenomena) which include climatic factors of light, temperature, precipitation, wind, topography, and soil (edaphic)

b. **Biotic component** include all living organisms present in the environmental system which can be grouped into two component

- Autotrophic component: include all green plants which fix the radiant energy of sun and manufactured food from inorganic substances
- Heterotrophic component: includes non-green plants and all animals which feed directly on autotrophs.

Although ecosystems vary from tidal pools and barrier reefs to arid stretches of scrub grass, they all share features that have been discovered as ecology, once an almost completely descriptive branch of biology, moved in an increasingly experimental direction. These features are:

- Energy flow
- Nutrient cycling
- Regulation of population size (numbers of individuals)

Energy Flow

Energy moves through the community of an ecosystem in a single direction by means of a food chain (web) in which there are the eaters, the eaten, and a combination of both. Populations are assigned an "occupational" role in an ecosystem in terms of their relation to the overall flow of this energy (food) in the food chain:

Producers: the first group in a food chain, usually consisting of green plants, which convert some of the energy of the sun (through photosynthesis) into organic molecules they use and store in their tissues.

Consumers: animals that feed on green plants and each other. *Primary* consumers are *herbivores*, which subsist on the primary producing plants. *Secondary* consumers feed on primary consumers, while *tertiary*, *quaternary*, etc., consumers are further along the chain.

Decomposers: bacteria, fungi, plants, or animals that feed on dead organisms and release the bound organic material of the organisms to the food chain.

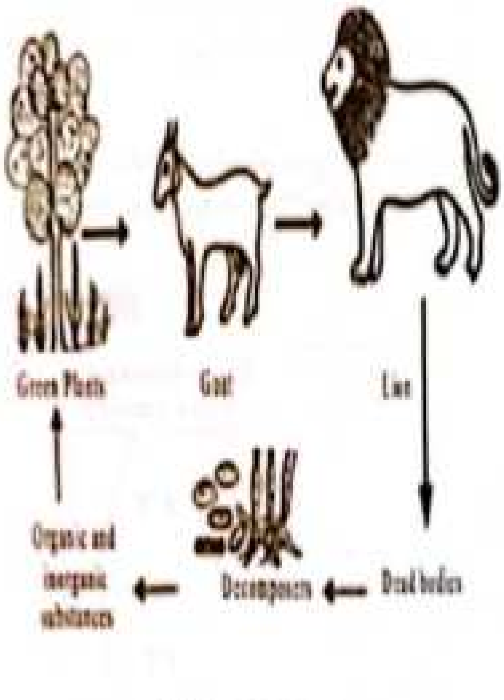


Fig. Food chain in a forest

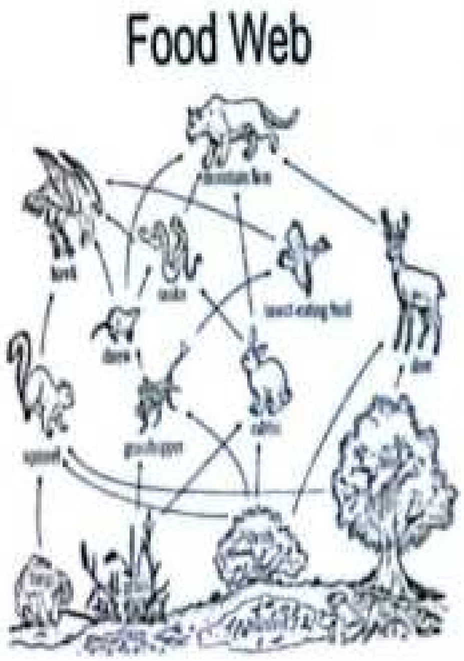


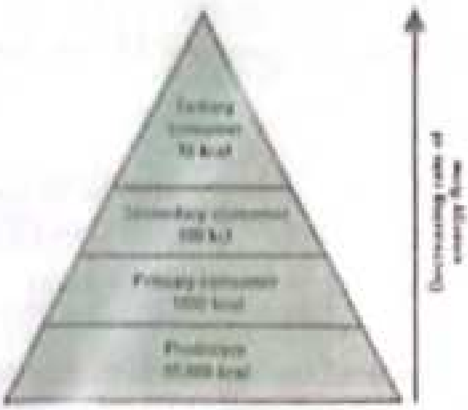
Figure 1: Food chain and Food web

Ecological pyramids

Energy can pass from one trophic level to the next when organic molecules from an organism's body are eaten by another organism.. There are three types of pyramids and see how they reflect the structure and function of ecosystems.

- **Energy pyramids**

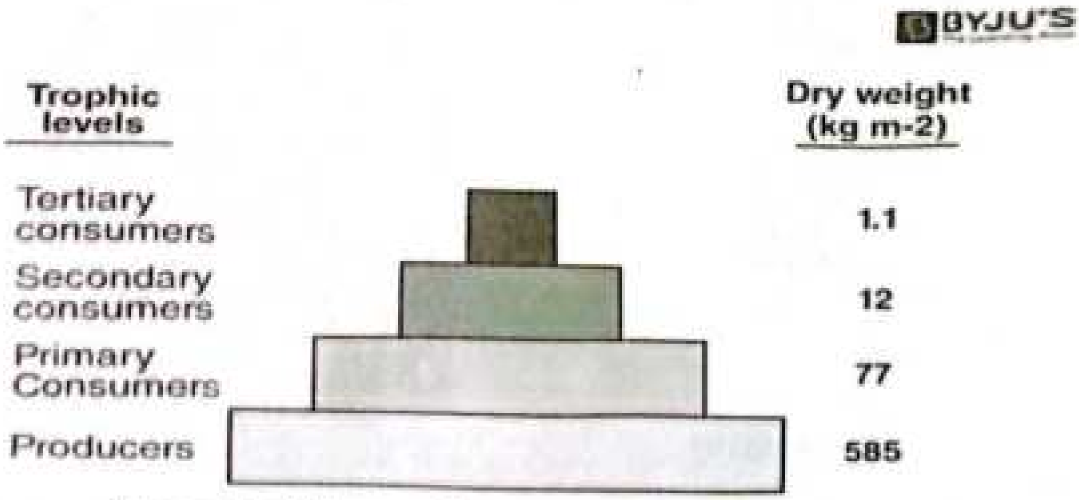
Energy pyramids represent energy flow through trophic levels. For instance, the pyramid below shows gross productivity for each trophic level in the Silver Springs ecosystem. An energy pyramid usually shows *rates* of energy flow through trophic levels, *not absolute* amounts of energy stored. It can have energy units, such as kcal/m^2/yrs biomass units, g/m^2/y



Pyramid of Energy

- **Biomass pyramids**

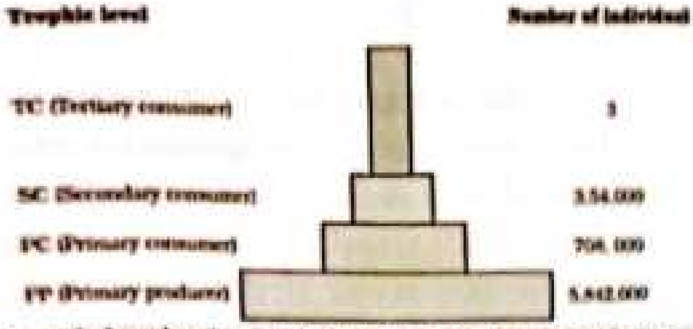
Another way to visualize ecosystem structure is with biomass pyramids. These pyramids represent the amount of energy that's stored in living tissue at the different trophic levels.



Pyramid of biomass for grassland ecosystem

- **Numbers pyramids**

Numbers pyramids show how many individual organisms there are in each trophic level. They can be upright, inverted, or kind of lumpy, depending on the ecosystem.



Pyramid of numbers in a grassland ecosystem. Only three top-carnivores are supported in an ecosystem based on production of nearly 6 millions plants

Nutrient Cycling

As energy in a food chain is passed along from one link to another, its useful capacity for work is diminished in accordance with the second law of thermodynamics. Heat is given off with every transformation. At the end of the food chain little or no free energy is left, so that recycling is not possible. On the other hand, matter is not lost as it passes from one component of the food chain to another-the passage of organic molecules and their elemental units along the food chain may be described in terms of a cycle. Generally, ecologists follow specific atoms through the cycle, e.g., carbon (C), nitrogen (N), and sulphur *(S)*, and chart their fates as they pass through the food chain, into the environment, and back again into the community.

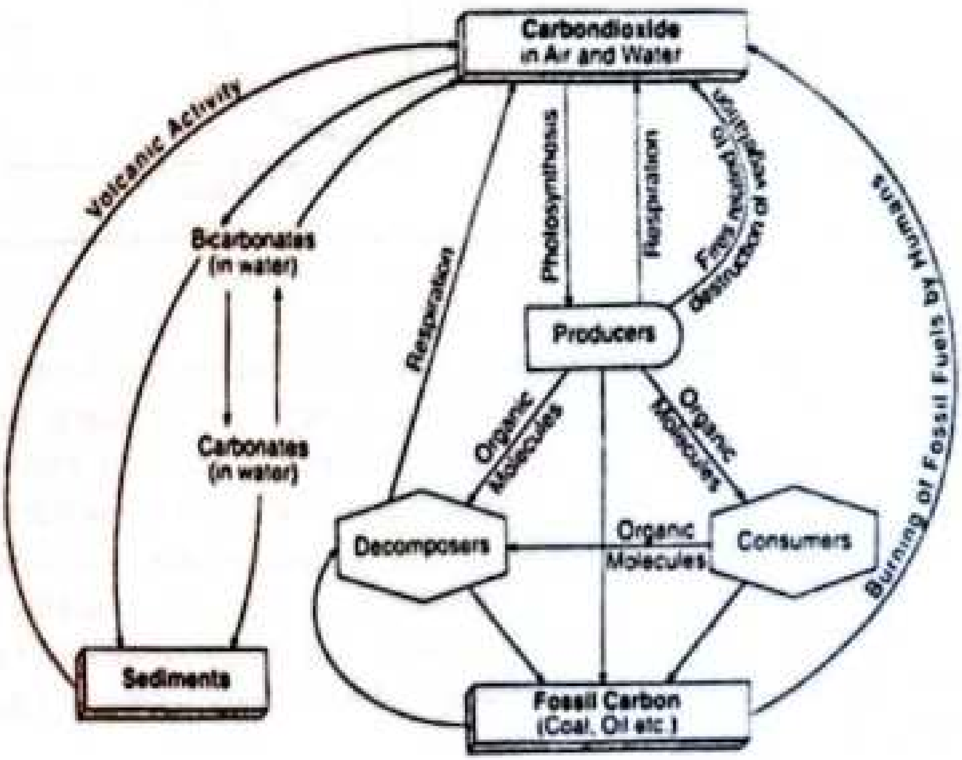


Figure.2: Carbon Cycle

The carbon cycle is powered alternately by the reduction (through photosynthesis) and the oxidation of carbon. Carbon enters the cycle as atmospheric CO_2, which is converted by plants to food during photosynthesis. Ultimately, this carbon is converted back to CO_2 through either respiration or combustion, and the cycle begins again.

Regulation of Population Size

Malthus and Darwin well knew, natural populations tend to increase exponentially, rather than incrementally by addition of a constant amount. Because of their high reproductive potential, such populations ***tend*** to double, then double again, and so on. A constant rate (reproductive potential) produces dramatic increases over time because as the population increases, the rate is being multiplied by an ever-increasing base value (the number of individuals in the population). The situation is summed up in the differential equation $dN/dt = rN$, where N is the number of individuals, t is time, and r is the intrinsic rate of increase. The left side of the equation, the change in number of individuals over time, is the rate of growth. This rate is equal to the intrinsic rate of increase r multiplied by the numbers present, since each individual shares in the tendency to increase in numbers.

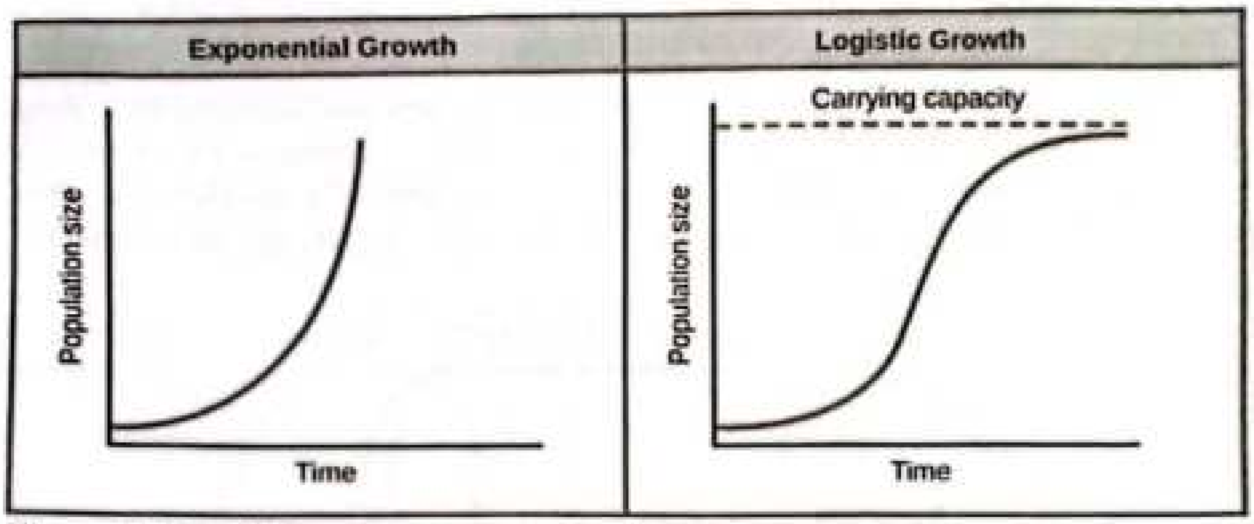


Figure.2: Growth Curves

Such a situation produces the typical exponential growth curve seen in Fig 2. Growth increases slowly at first and then rapidly as the steady rate is applied to increasing numbers of individuals. The slope may even approach a vertical line. But such a depiction is only hypothetical, since a population that increases in size at such a great rate will be subject to limiting constraints. Its birthrate will soon be matched or even overtaken by an increase in its death rate, which will tend to maintain the population at a steady number. Stable population size is usually characterized by a modification of the previous equation as follows: $dN/dt = rN[(K - N)/K]$, where K is the ***carrying capacity***, or the maximum number of individuals a habitat can support.

Here the rate of growth is modified by a factor $(K - N)/K$.